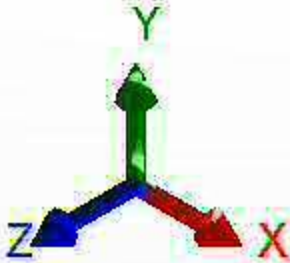
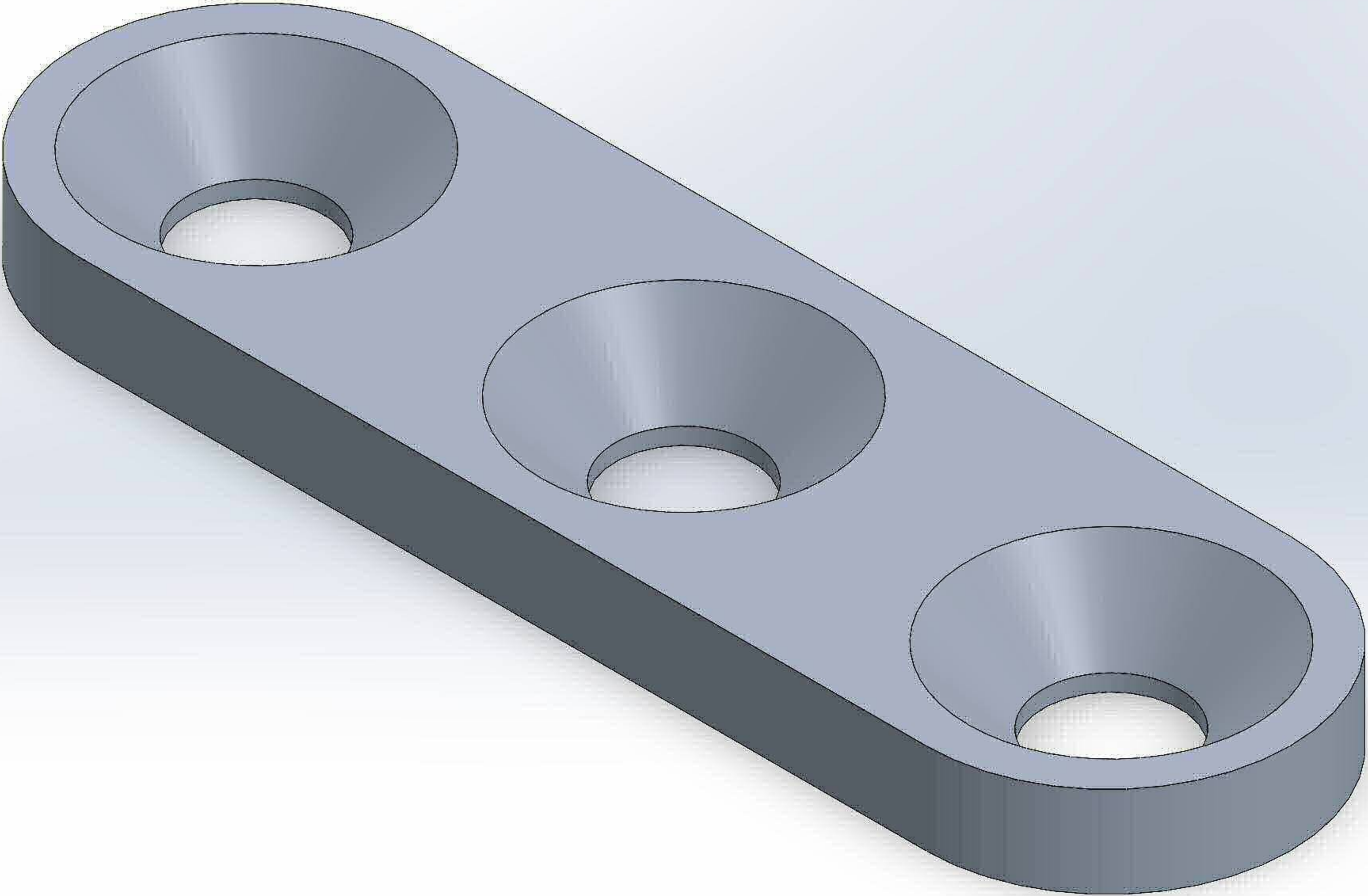


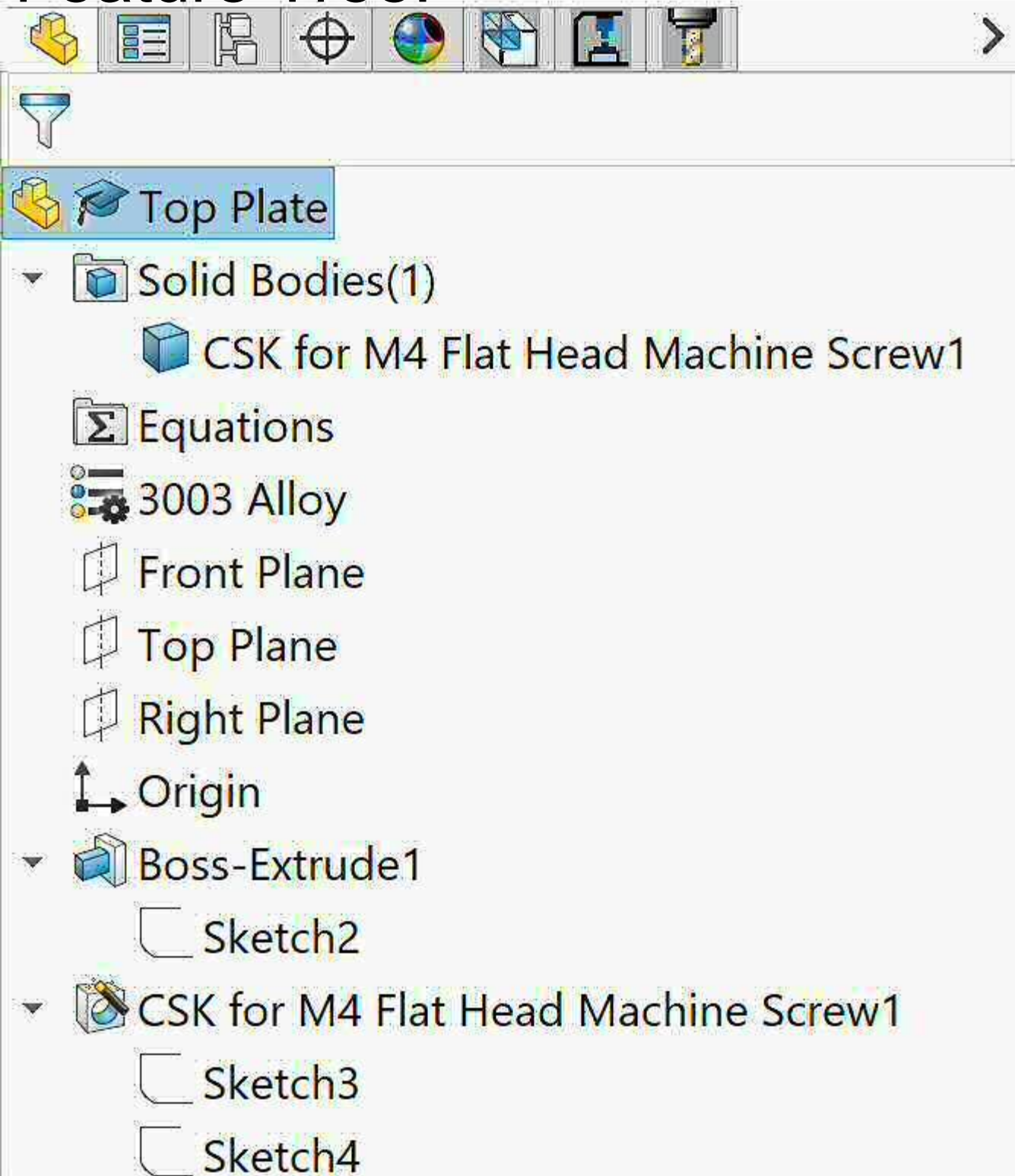
Isometric View:



*Isometric



Feature Tree:



Top Plate

Equations

3003 Alloy

Front Plane

Top Plane

Right Plane

Origin

Boss-Extrude1

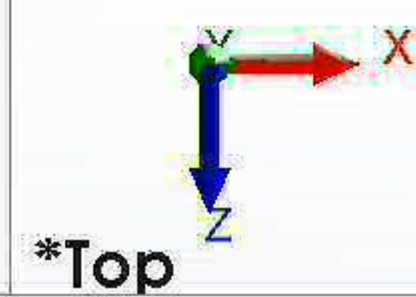
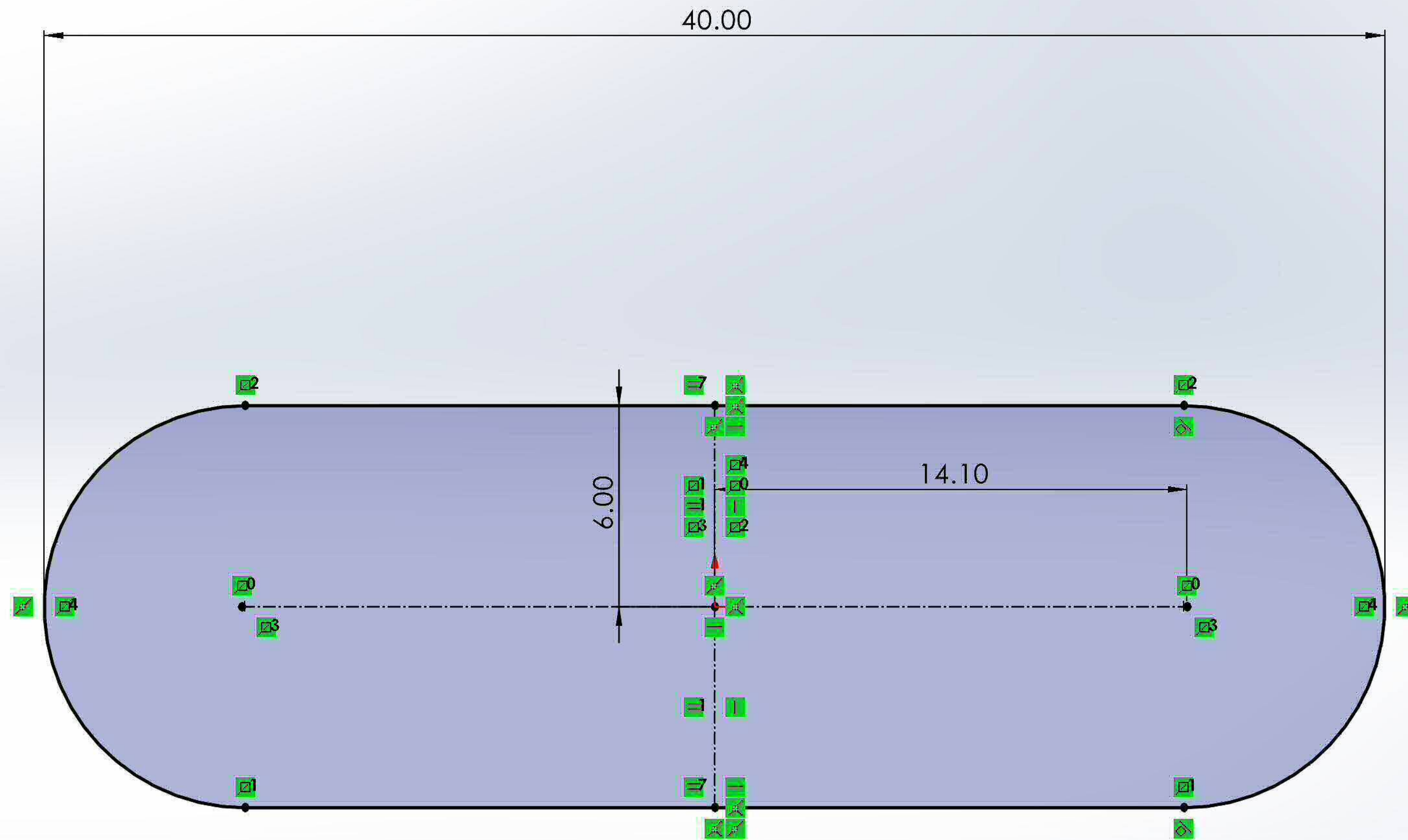
Sketch2

CSK for M4 Flat Head Machine Screw1

Sketch3

Sketch4

Sketch 2:





Hole Specification



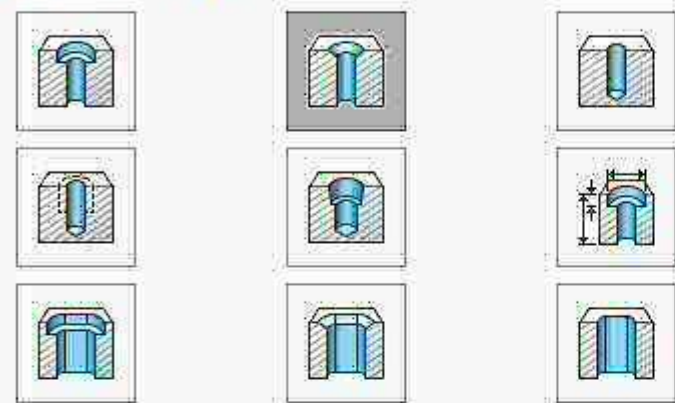
Hole Wizard:

Type Positions



No Favorite Selected

Hole Type



Standard:

ANSI Metric

Type:

Flat Head Screw - ANSI B18.6.7M

Hole Specifications

Size:

M4

Fit:

Normal

☒ Show custom sizing



4.500mm



9.400mm



90deg

End Condition



Through All

Options

Tolerance/Precision

Callout value

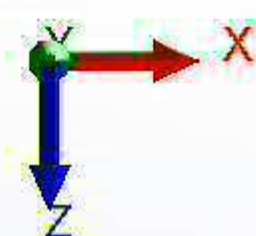
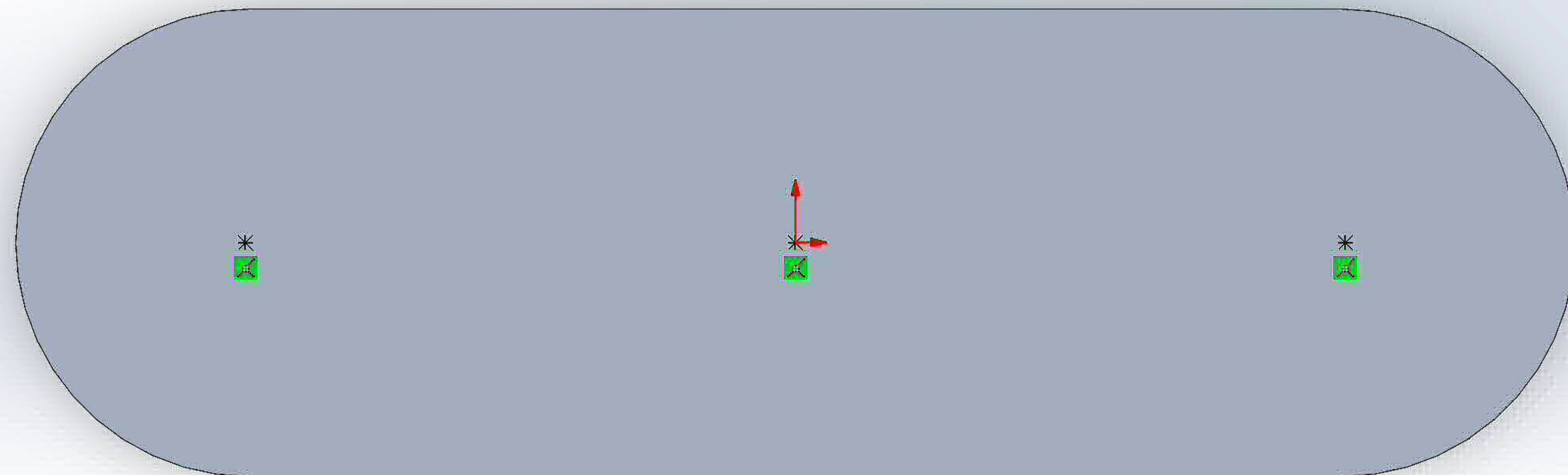
Thru Hole Diameter (4.5)



None



.12 (Document)



Top Plate

Solid Bodies(1)

Equations

3003 Alloy

Front Plane

Top Plane

Right Plane

Origin

Boss-Extrude1

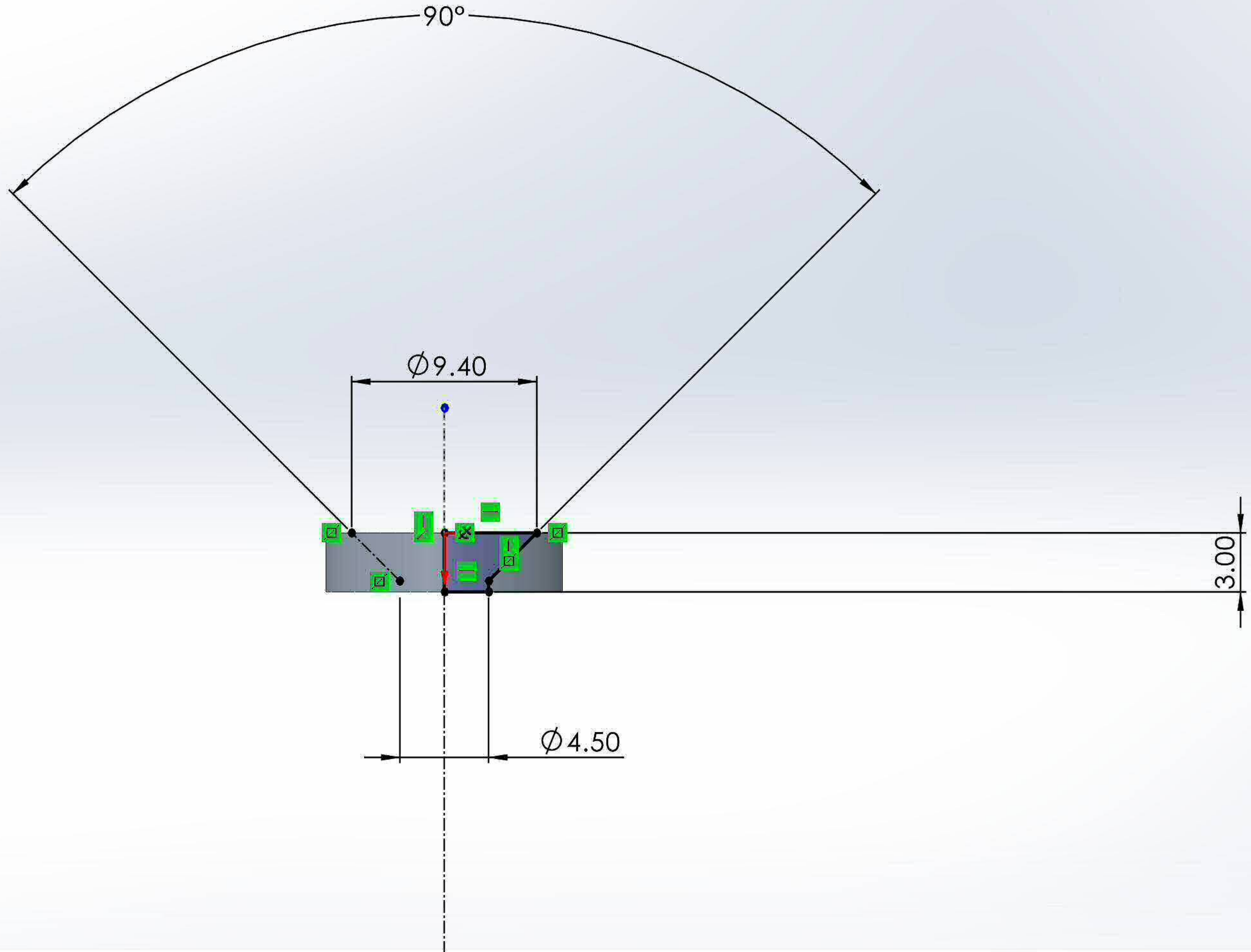
Sketch2

CSK for M4 Flat Head Machine Screw1

Sketch3

Sketch4

Sketch 4: (Created By The Hole Wizard)



*Left

Units Used:

MKS (meter, kilogram, second)

CGS (centimeter, gram, second)

MMGS (millimeter, gram, second)

IPS (inch, pound, second)

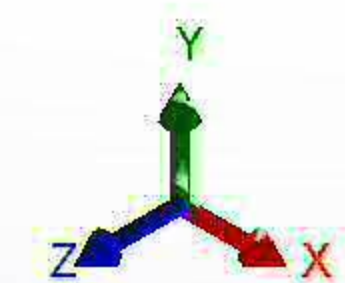
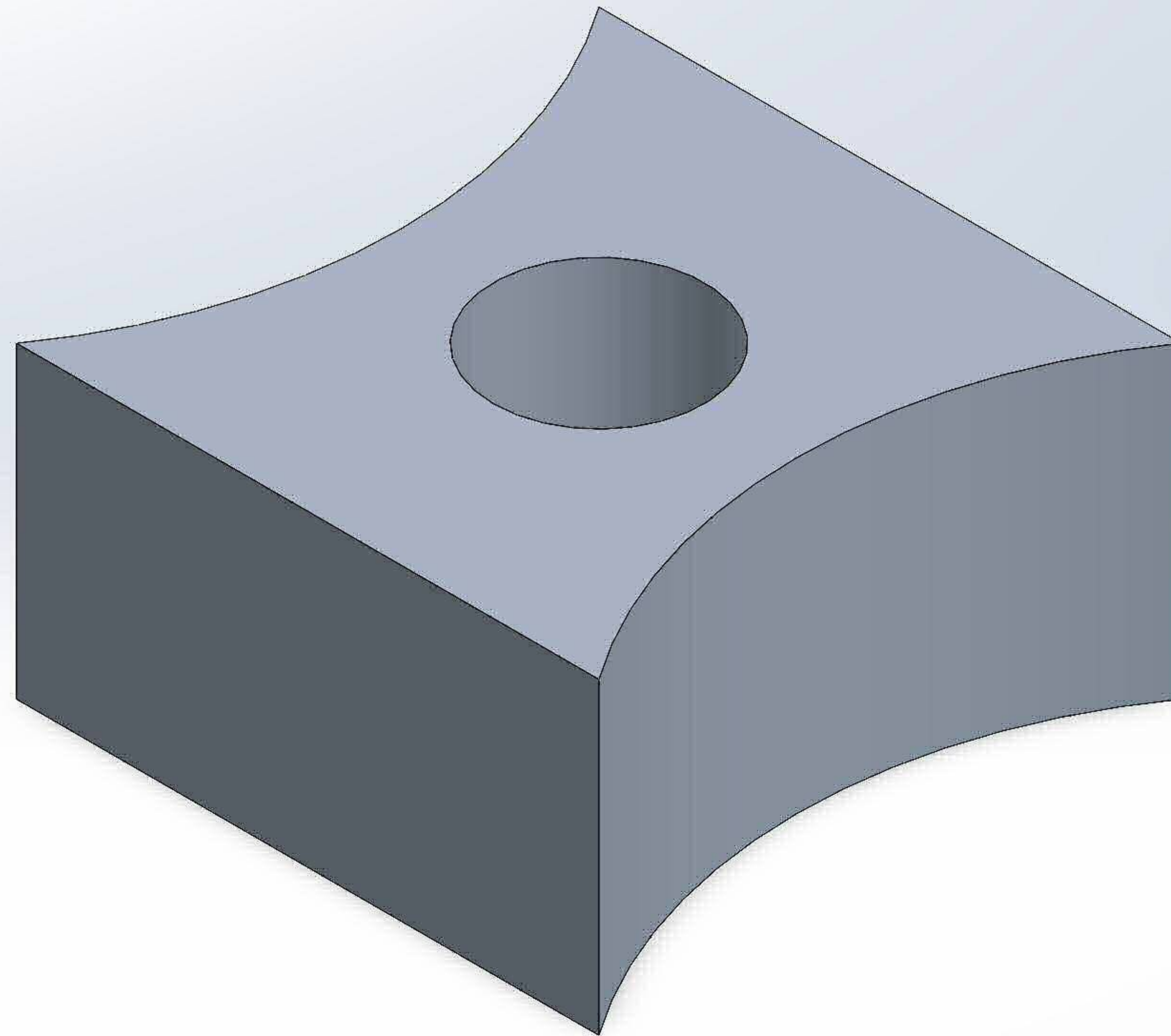
Edit Document Units...

Simplified Interface

MMGS



Isometric View:



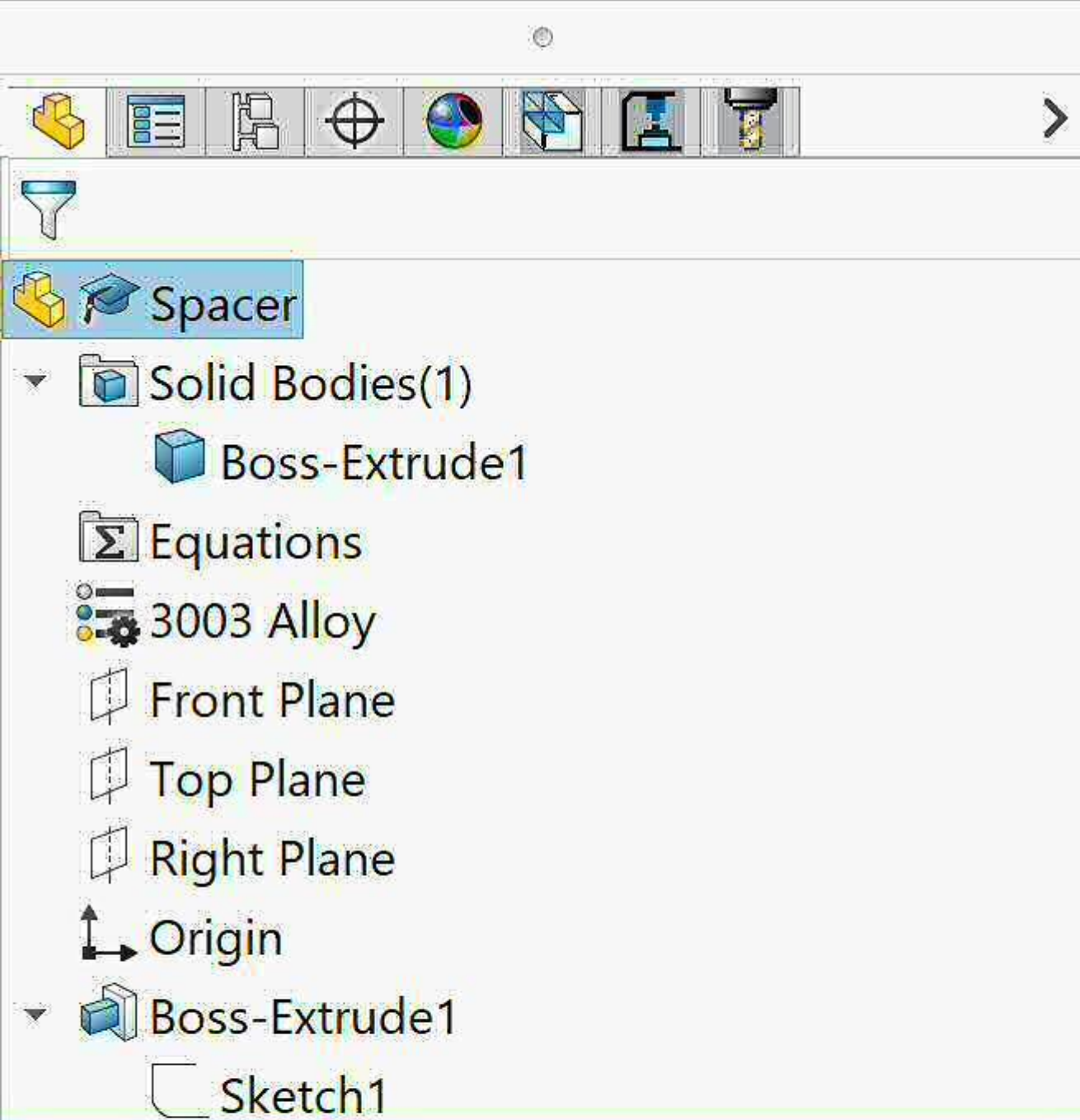
*Isometric

Editing Part

Simplified Interface

MMGS





Feature Tree:

Sketch 1:

Spacer

Equations

3003 Alloy

Front Plane

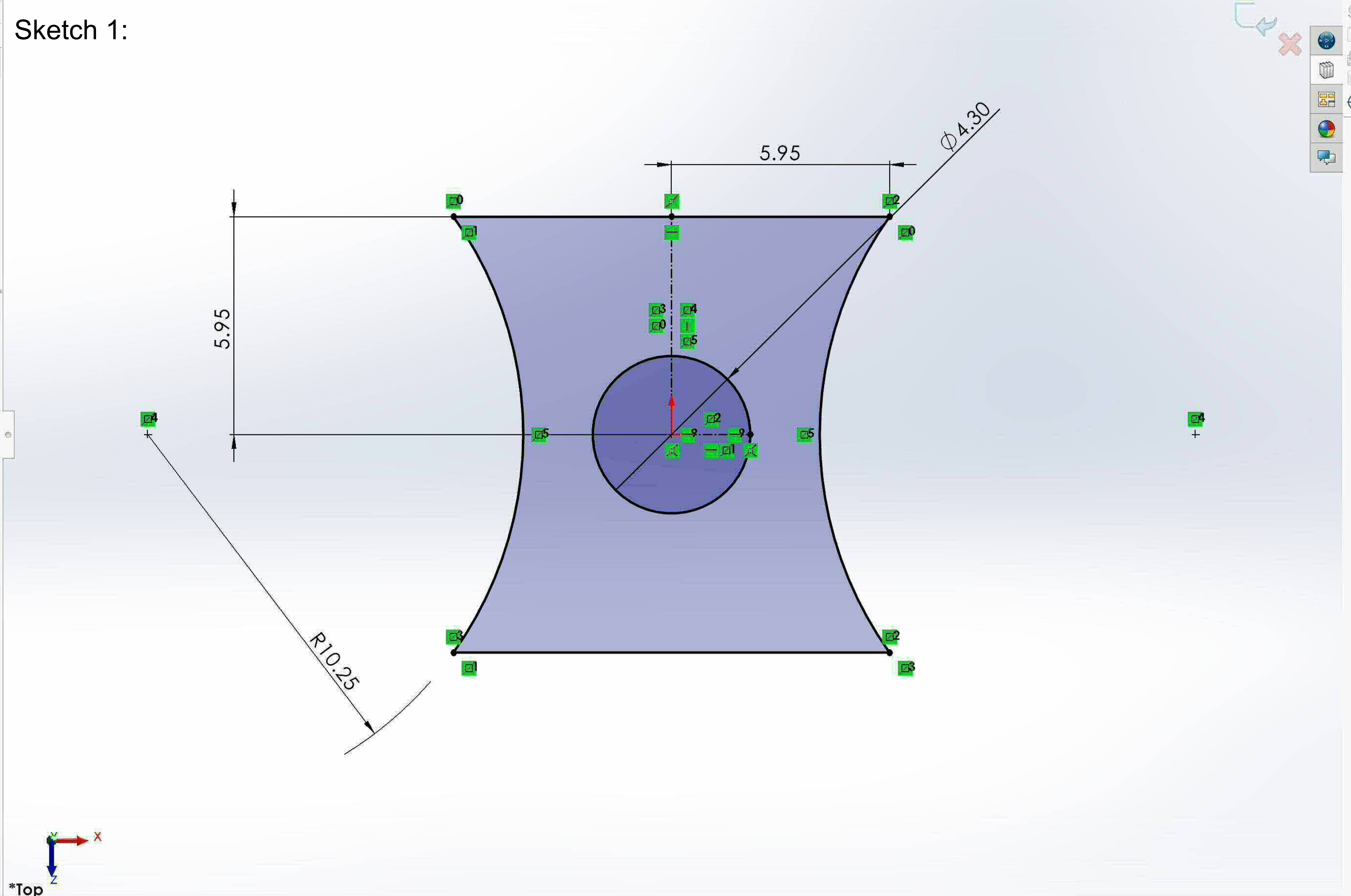
Top Plane

Right Plane

Origin

Boss-Extrude1

Sketch1





Mass Properties:



Spacer

Options...

Override Mass Properties...

Recalculate

☒ Include hidden bodies/components☐ Create Center of Mass feature☐ Show weld bead mass

Report coordinate values relative to: -- default --

Mass properties of Spacer

Configuration: Default

Coordinate system: -- default --

Density = 0.00 grams per cubic millimeter

Mass = 1.64 grams

Volume = 606.52 cubic millimeters

Surface area = 587.56 square millimeters

Center of mass: (millimeters)

X = 0.00

Y = 0.00

Z = 0.00

Principal axes of inertia and principal moments of inertia: (grams * square millimeters)

Taken at the center of mass.

I_x = (0.00, 0.00, 1.00) P_x = 19.35I_y = (1.00, 0.00, 0.00) P_y = 29.76I_z = (0.00, 1.00, 0.00) P_z = 38.27

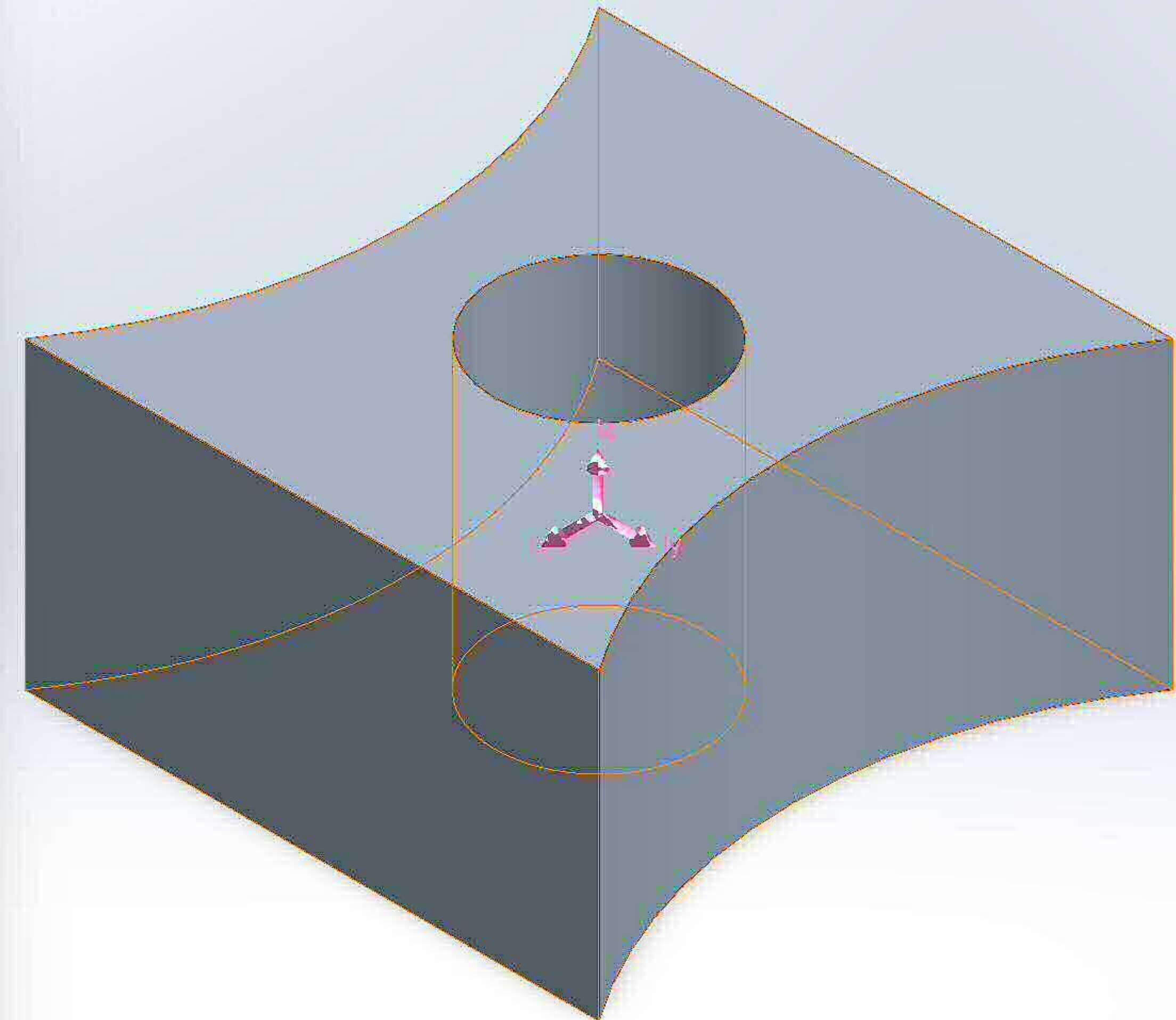
Moments of inertia: (grams * square millimeters)

Taken at the center of mass and aligned with the output coordinate system. (Using positive tensor notation.)

L_{xx} = 29.76 L_{xy} = 0.00 L_{xz} = 0.00L_{yx} = 0.00 L_{yy} = 38.27 L_{yz} = 0.00L_{zx} = 0.00 L_{zy} = 0.00 L_{zz} = 19.35

Moments of inertia: (grams * square millimeters)

Taken at the output coordinate system. (Using positive tensor notation.)

I_{xx} = 29.76 I_{xy} = 0.00 I_{xz} = 0.00I_{yx} = 0.00 I_{yy} = 38.27 I_{yz} = 0.00I_{zx} = 0.00 I_{zy} = 0.00 I_{zz} = 19.35

Units Used:

MKS (meter, kilogram, second)

CGS (centimeter, gram, second)

MMGS (millimeter, gram, second)

IPS (inch, pound, second)

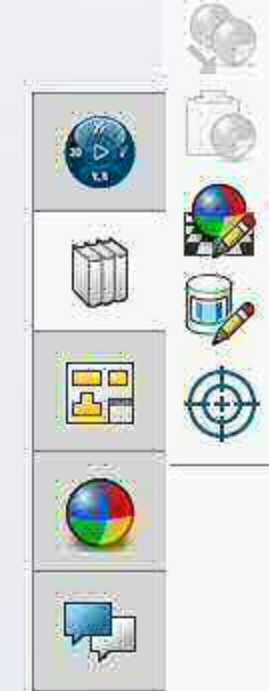
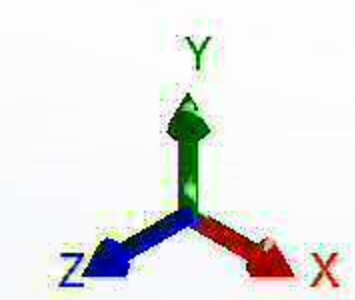
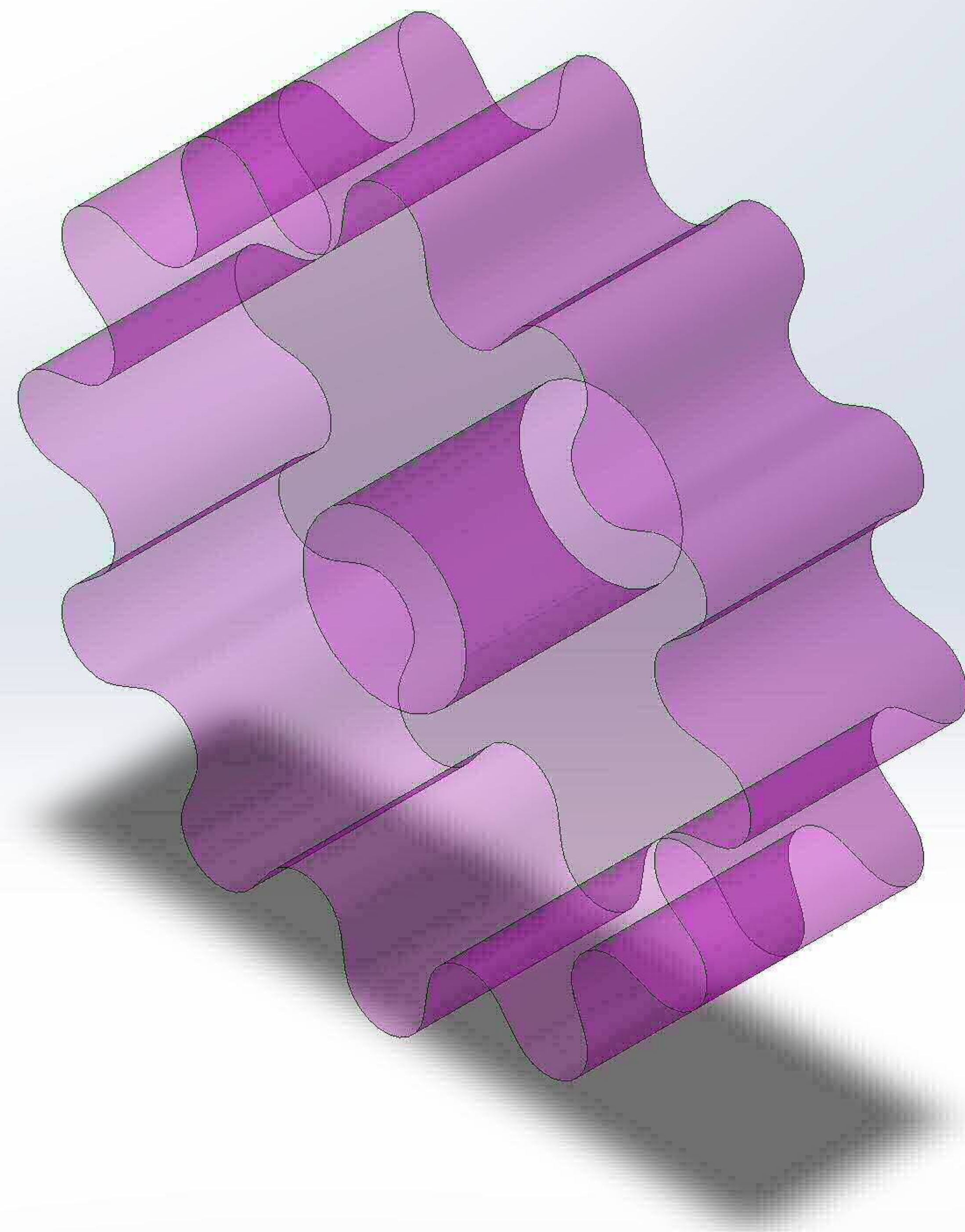
Edit Document Units...

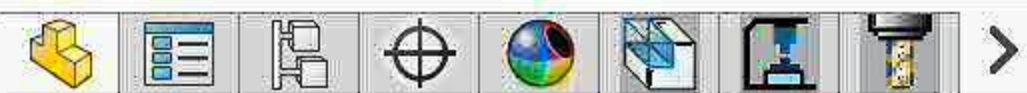
Simplified Interface

MMGS



Isometric View:





Feature Tree:

  Spur Gear

▼  Solid Bodies(1)

 Cut-Extrude1

 Equations

 Acrylic (Medium-high impact)

 Front Plane

 Top Plane

 Right Plane

 Origin

▼  Boss-Extrude1

 Sketch2

▼  Cut-Extrude1

 Sketch3

Sketch 2:

Spur Gear

Solid Bodies(1)

- Cut-Extrude1

Equations

Acrylic (Medium-high impact)

Front Plane

Top Plane

Right Plane

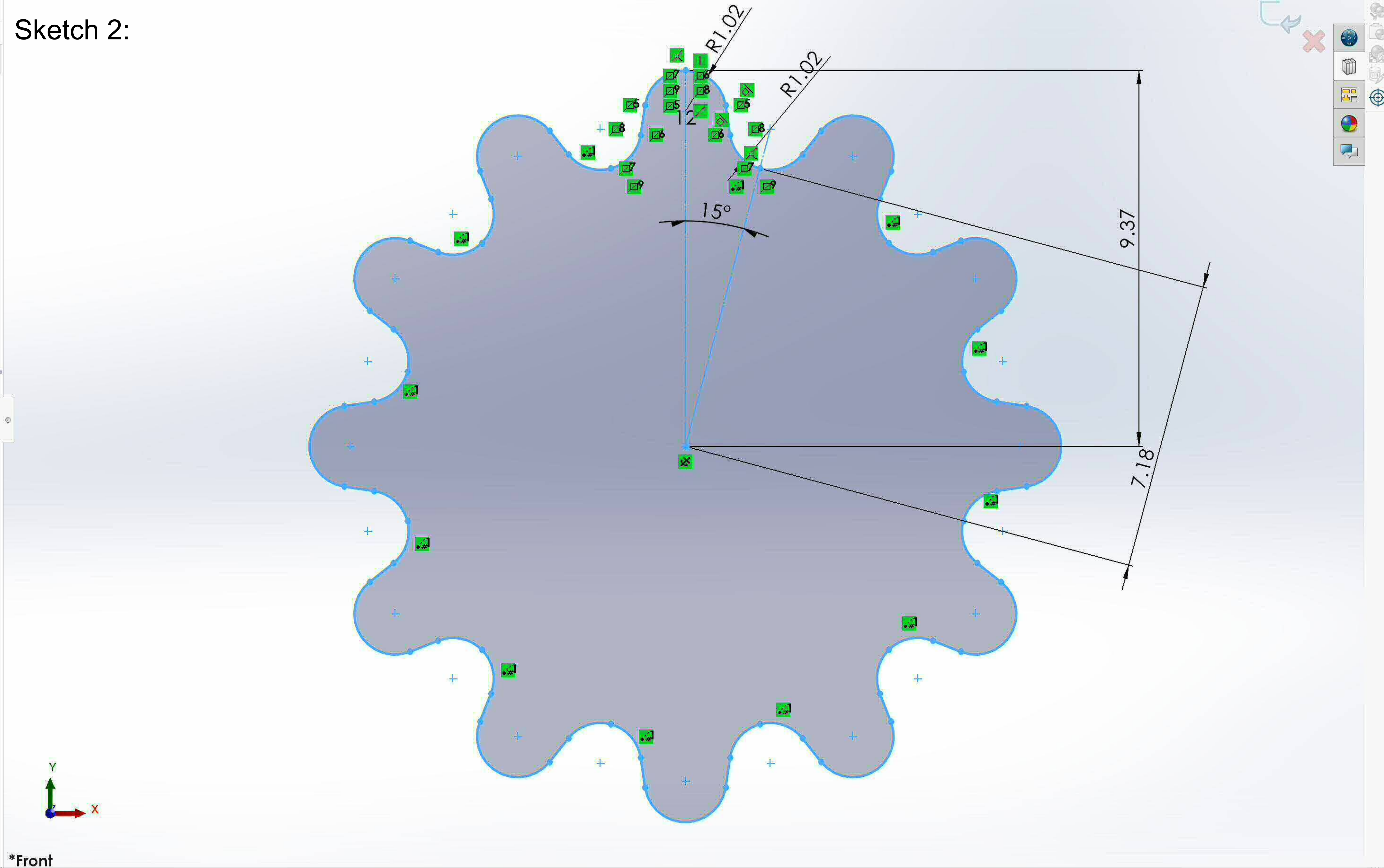
Origin

Boss-Extrude1

Sketch2

Cut-Extrude1

Sketch3



Sketch 3:

Spur Gear

Solid Bodies(1)

- Boss-Extrude1

Equations

Acrylic (Medium-high impact)

Front Plane

Top Plane

Right Plane

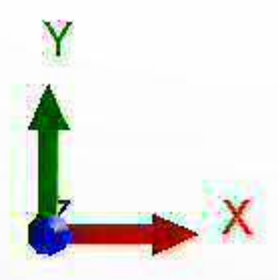
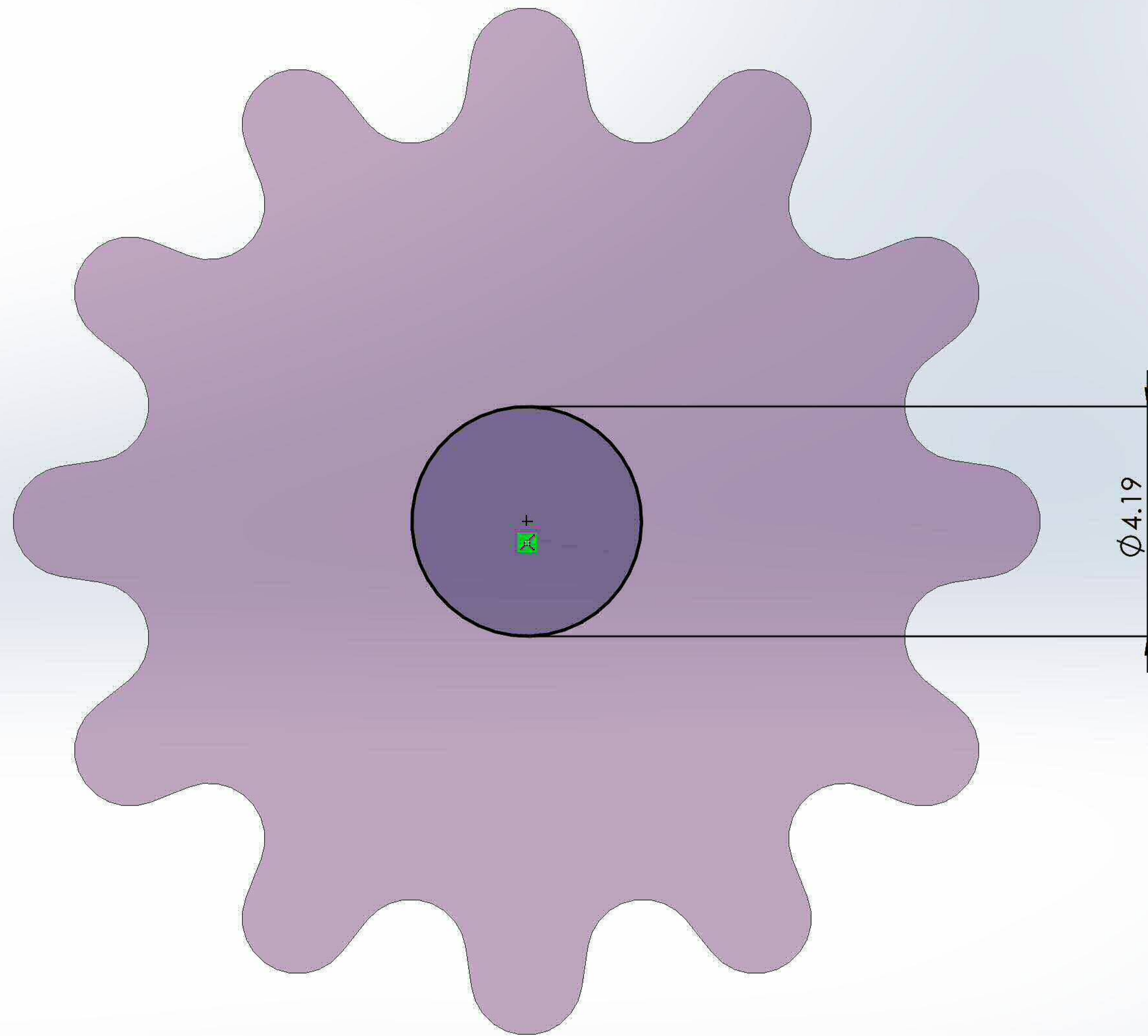
Origin

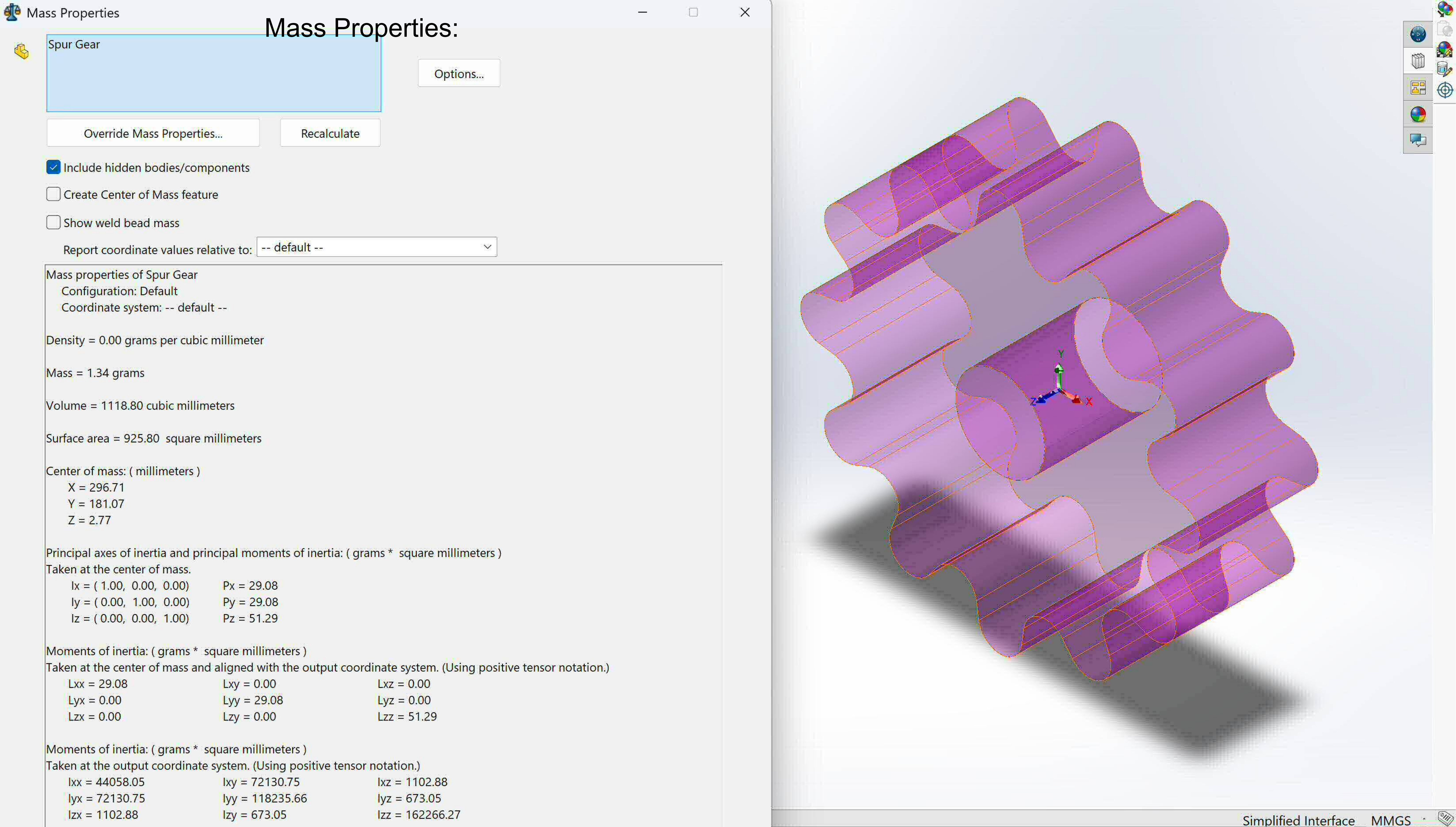
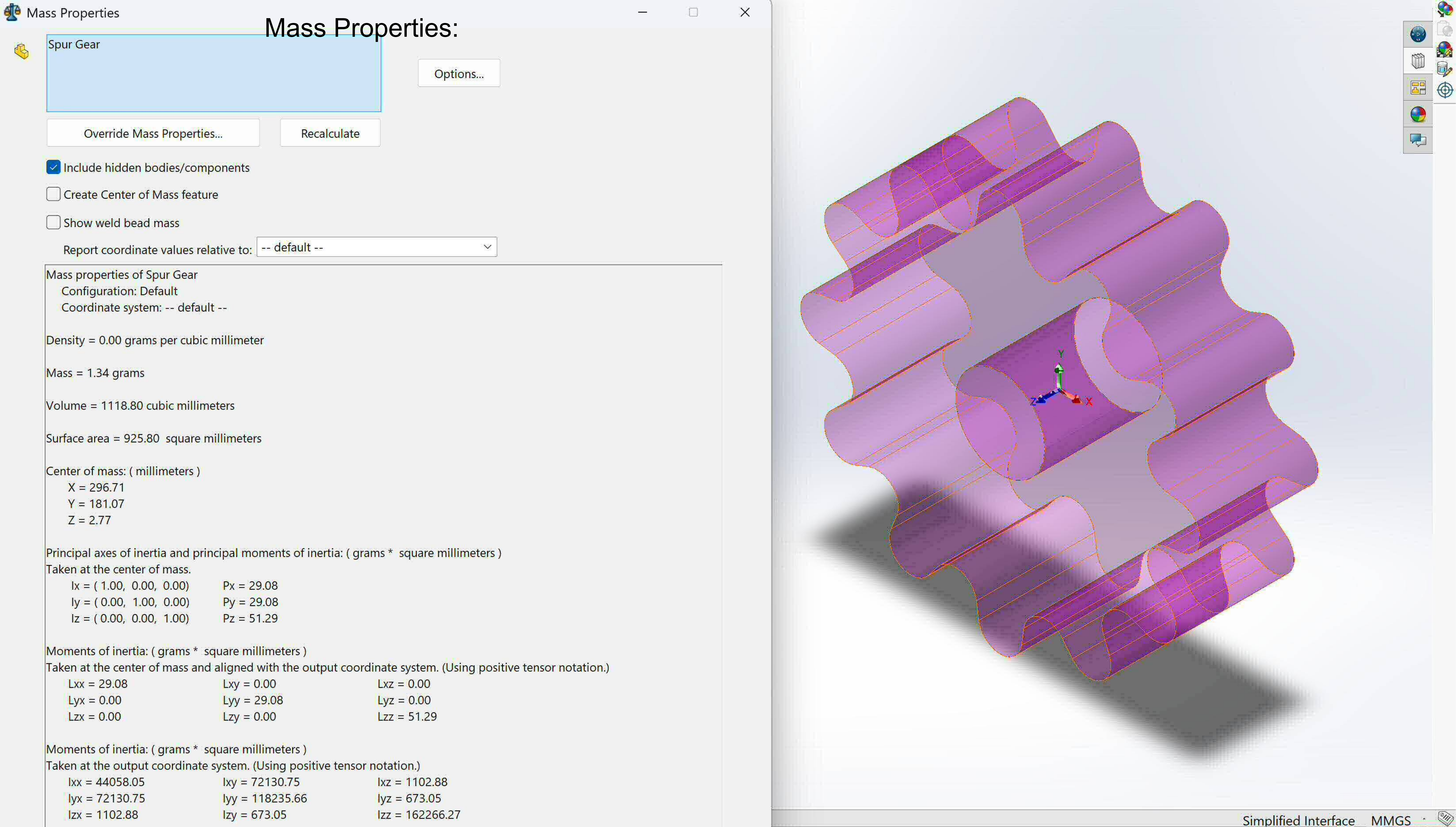
Boss-Extrude1

- Sketch2

Cut-Extrude1

- Sketch3





Units Used:

MKS (meter, kilogram, second)

CGS (centimeter, gram, second)

MMGS (millimeter, gram, second)

IPS (inch, pound, second)

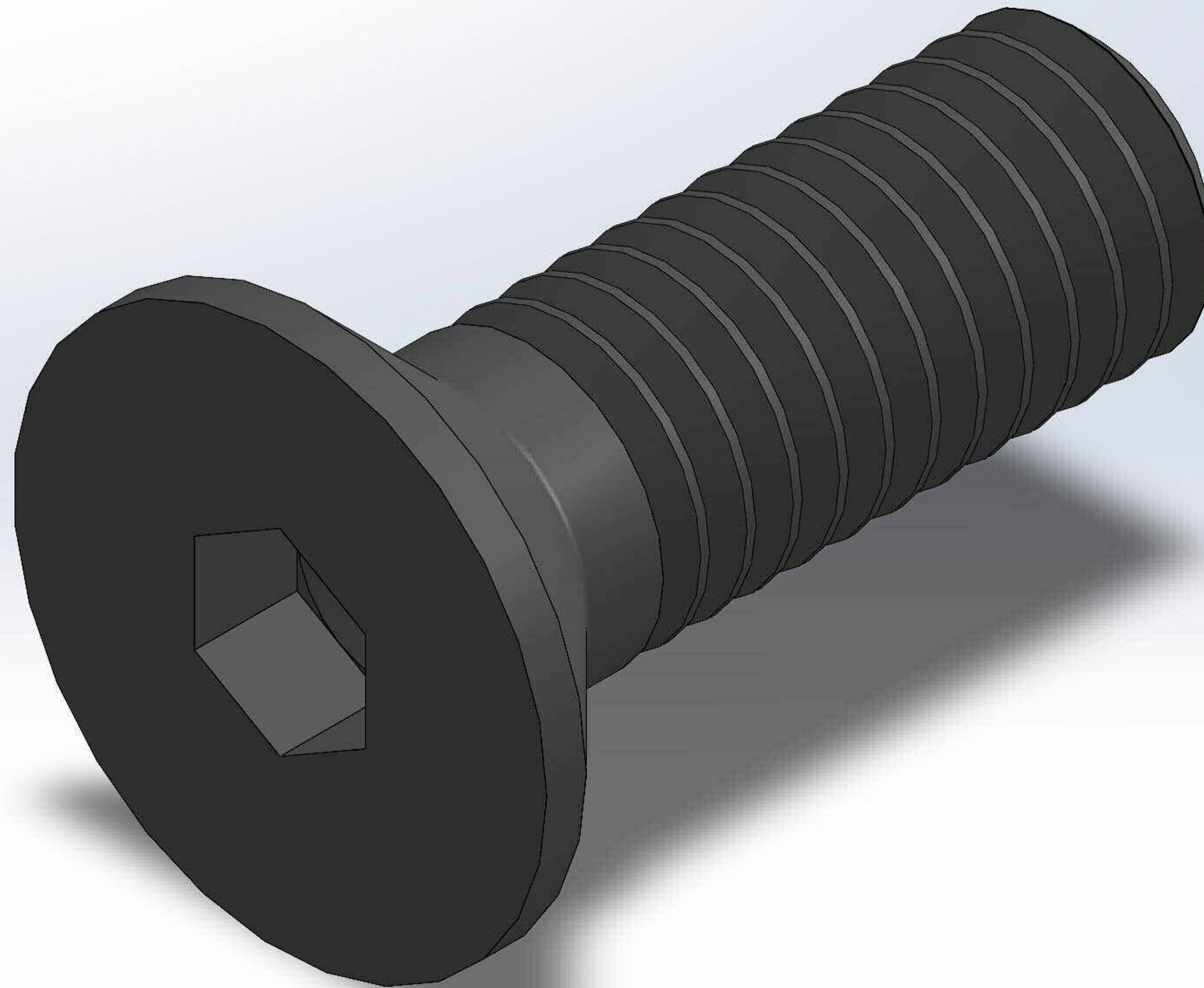
Edit Document Units...

Simplified Interface

MMGS



Isometric View:





Feature Tree:



M4 Screw

▼  Solid Bodies(1)



ThdSchPat



Equations



AISI 1020



Plane1



Plane2



Plane3



Origin

▼  MateReferences



MateReference

▼  BaseBody



BodySke



ThreadCosmetic



Fillet1

▼  PreBroach



Sketch2

▼  Hex



Sketch3

▼  ThreadSchematic



ThdSchSke



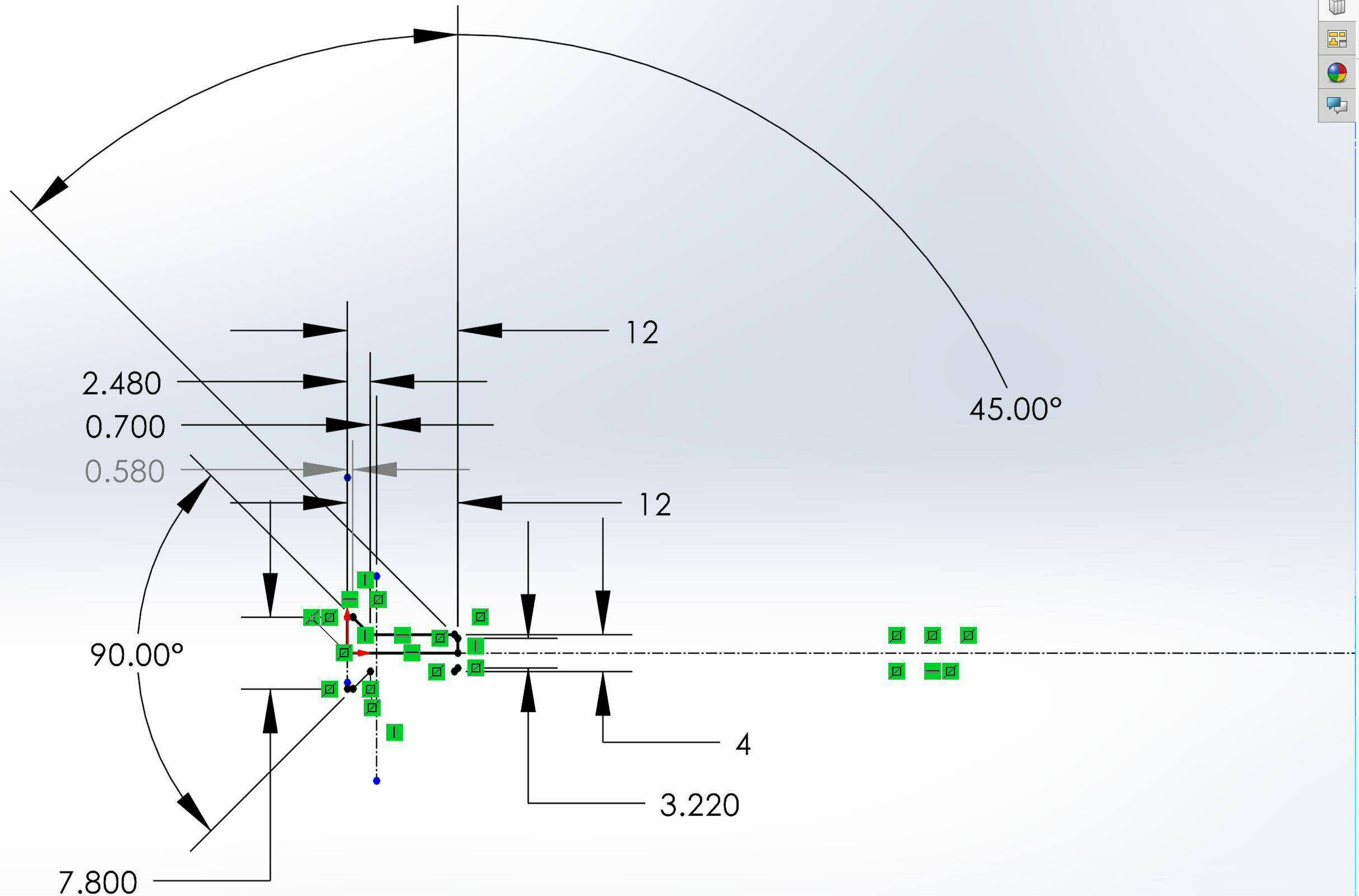
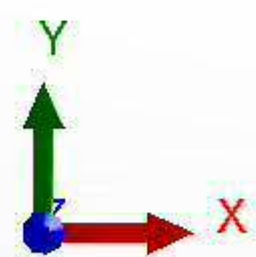
ThdSchPat

BodySke Sketch:

M4 Screw
Equations
AISI 1020
Plane1
Plane2
Plane3
Origin
MateReferences
MateReference
BaseBody
BodySke
ThreadCosmetic
Fillet1
PreBroach
Sketch2
Hex
Sketch3
ThreadSchematic
ThdSchSke
ThdSchPat

BodySke Sketch:

*Right



Sketch 2:

M4 Screw

Solid Bodies(1)

- Fillet1

Equations

AISI 1020

Plane1

Plane2

Plane3

Origin

MateReferences

- MateReference

BaseBody

- BodySke
- ThreadCosmetic

Fillet1

PreBroach

- Sketch2

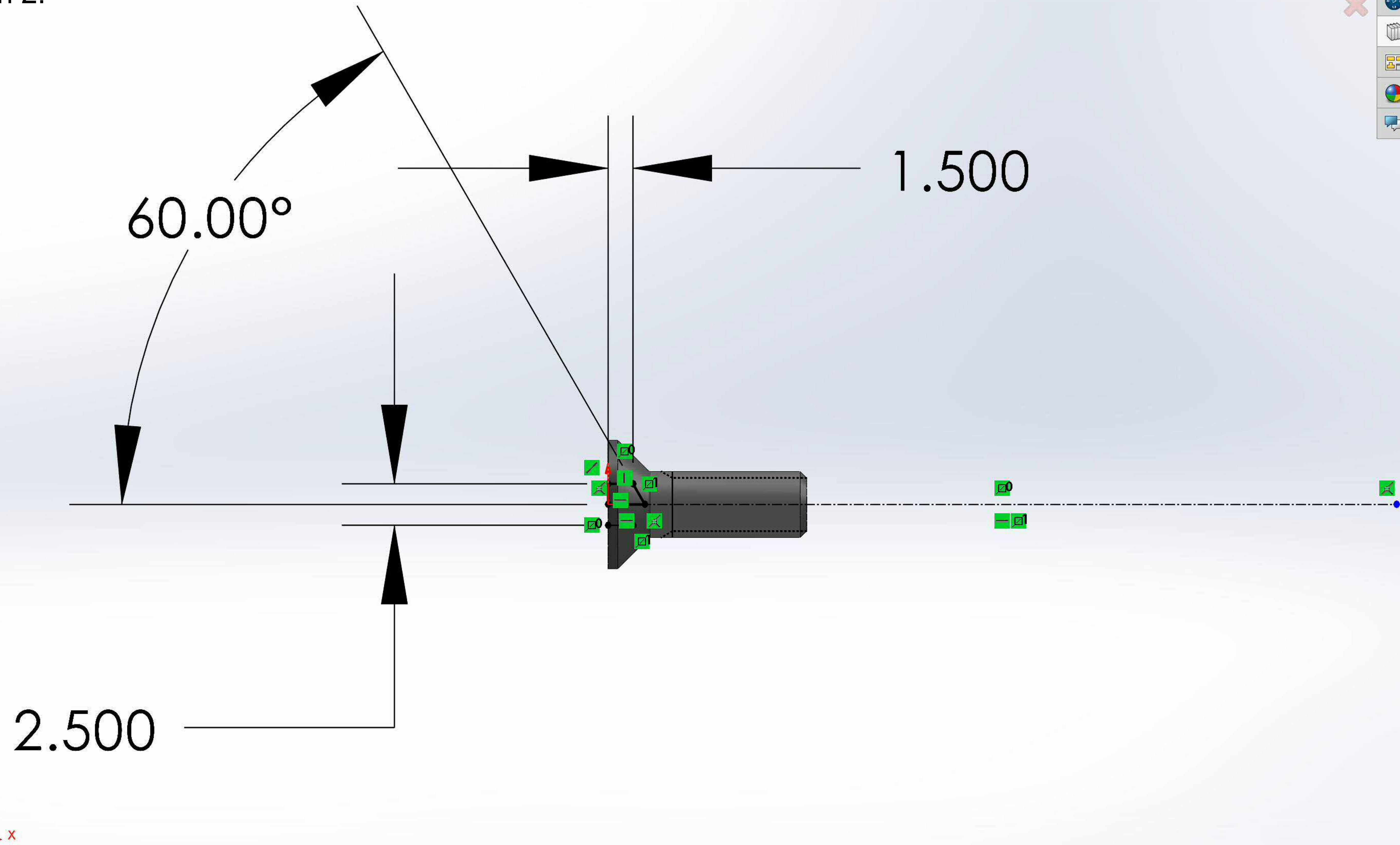
Hex

- Sketch3

ThreadSchematic

- ThdSchSke
- ThdSchPat

Sketch 2:



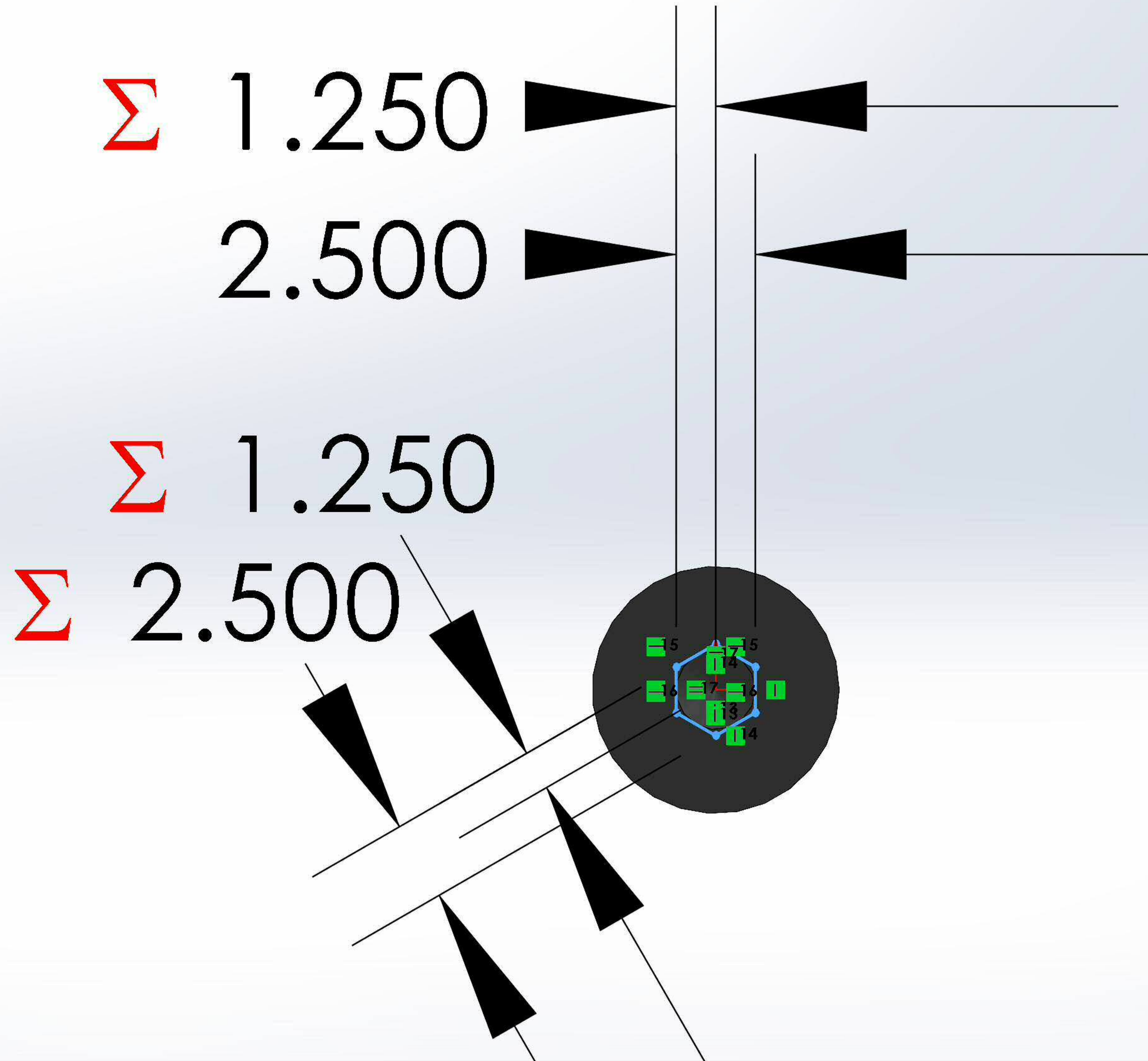
*Right

Sketch 3:

Navigation icons: Home, Feature Tree, Isolate, Hide, Rotate, Zoom, etc.

Feature Tree:

- M4 Screw
- Solid Bodies(1)
 - PreBroach
- Equations
- AISI 1020
- Plane1
- Plane2
- Plane3
- Origin
- MateReferences
 - MateReference
- BaseBody
 - BodySke
 - ThreadCosmetic
- Fillet1
- PreBroach
 - Sketch2
- Hex
 - Sketch3
- ThreadSchematic
 - ThdSchSke
 - ThdSchPat



*Front

ThdSchSke Sketch:

M4 Screw

Solid Bodies(1)

- Hex

Equations

AISI 1020

Plane1

Plane2

Plane3

Origin

MateReferences

- MateReference

BaseBody

- BodySke
- ThreadCosmetic

Fillet1

PreBroach

- Sketch2

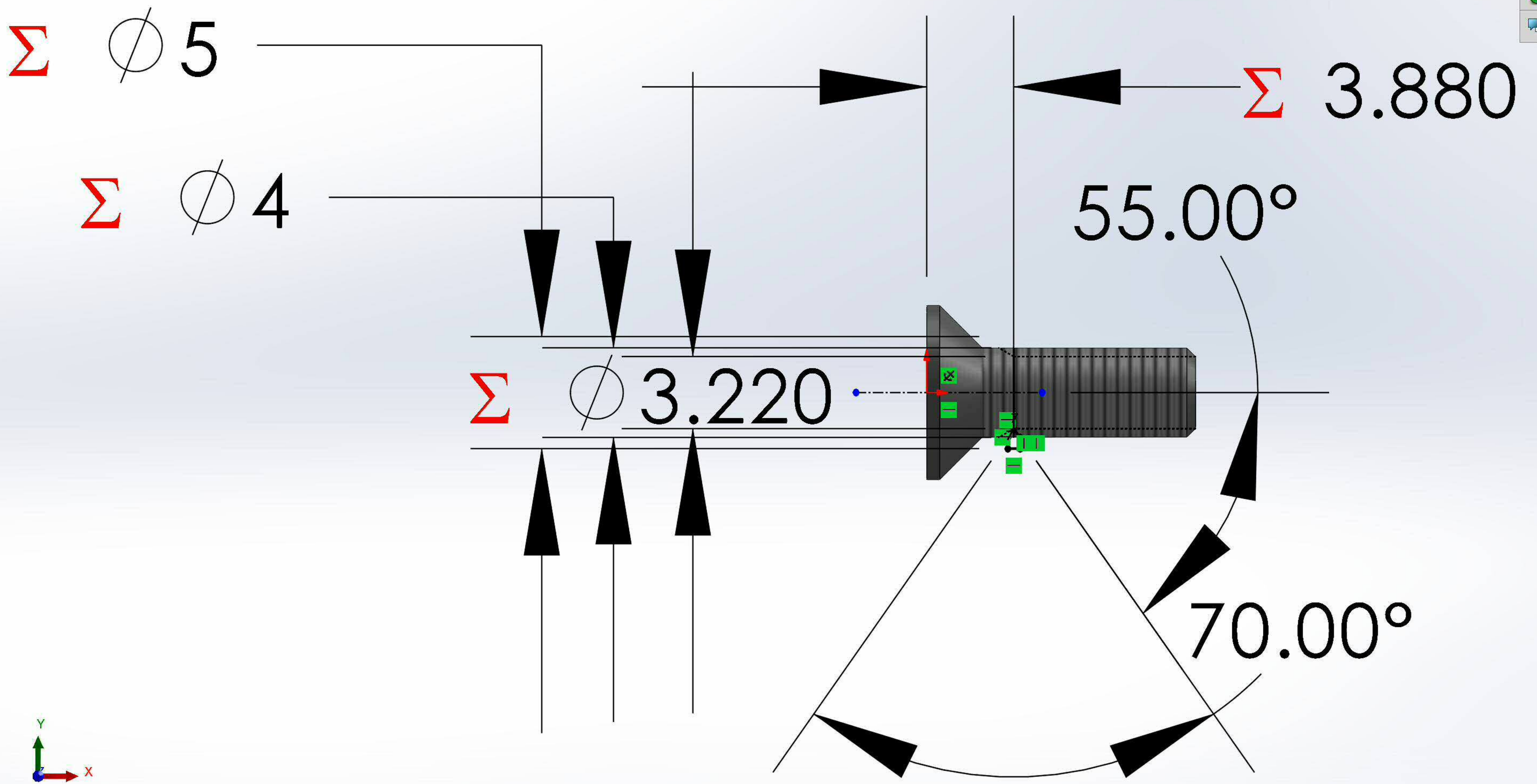
Hex

- Sketch3

ThreadSchematic

- ThdSchSke
- ThdSchPat

ThdSchSke Sketch:



*Right



Mass Properties:



M4 Screw

Options...

Override Mass Properties...

Recalculate

☒ Include hidden bodies/components☐ Create Center of Mass feature☐ Show weld bead mass

Report coordinate values relative to: -- default --

Mass properties of M4 Screw

Configuration: B18.3.5M - 4 x 0.7 x 12 Socket FCHS -- 12C

Coordinate system: -- default --

Density = 0.008 grams per cubic millimeter

Mass = 1.390 grams

Volume = 175.889 cubic millimeters

Surface area = 301.806 square millimeters

Center of mass: (millimeters)

X = 4.622

Y = 0.000

Z = 0.000

Principal axes of inertia and principal moments of inertia: (grams * square millimeters)

Taken at the center of mass.

I_x = (1.000, 0.000, 0.000) P_x = 5.213I_y = (0.000, 0.000, -1.000) P_y = 21.917I_z = (0.000, 1.000, 0.000) P_z = 21.917

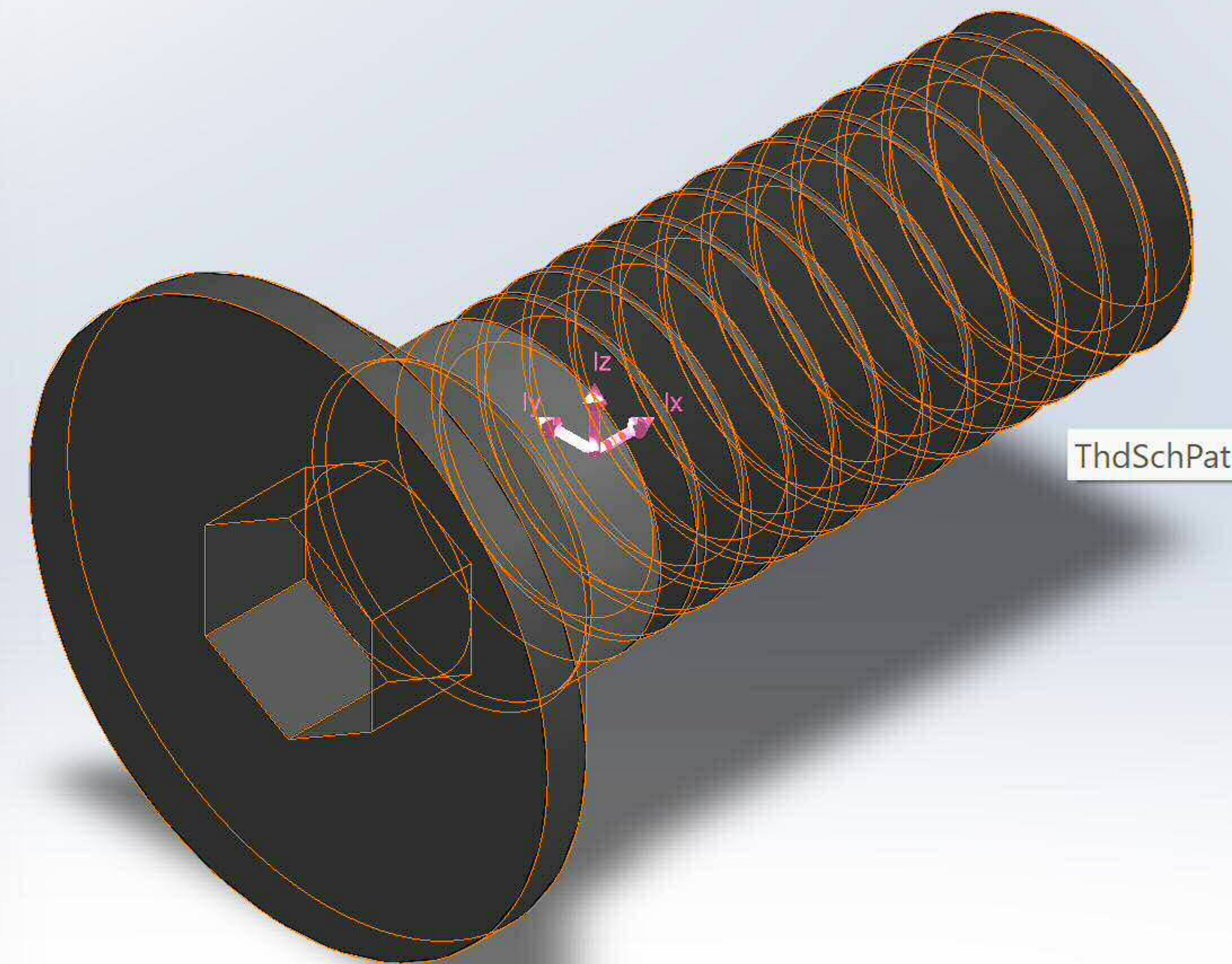
Moments of inertia: (grams * square millimeters)

Taken at the center of mass and aligned with the output coordinate system. (Using positive tensor notation.)

L_{xx} = 5.213 L_{xy} = 0.000 L_{xz} = 0.000L_{yx} = 0.000 L_{yy} = 21.917 L_{yz} = 0.000L_{zx} = 0.000 L_{zy} = 0.000 L_{zz} = 21.917

Moments of inertia: (grams * square millimeters)

Taken at the output coordinate system. (Using positive tensor notation.)

I_{xx} = 5.213 I_{xy} = 0.000 I_{xz} = 0.000I_{yx} = 0.000 I_{yy} = 51.605 I_{yz} = 0.000I_{zx} = 0.000 I_{zy} = 0.000 I_{zz} = 51.605

ThdSchPat

Units Used:

MKS (meter, kilogram, second)

CGS (centimeter, gram, second)

MMGS (millimeter, gram, second)

IPS (inch, pound, second)

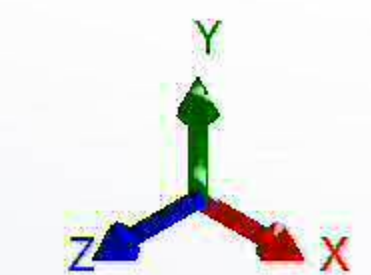
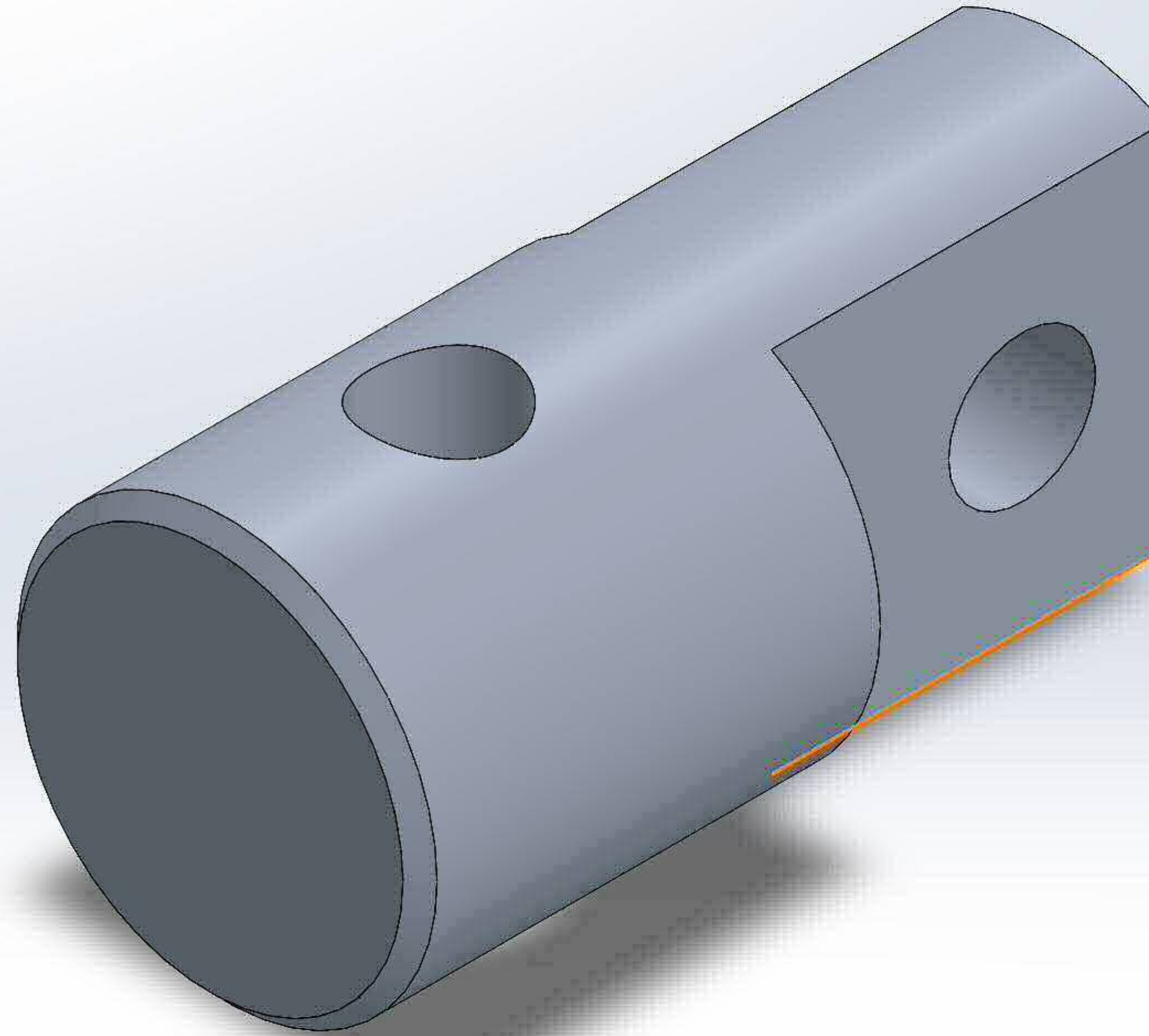
Edit Document Units...

Simplified Interface

MMGS



Isometric View:





Feature Tree:

  Keyring Connector

▼  Solid Bodies(1)
  Cut-Extrude2

 Equations

 3003 Alloy

 Front Plane

 Top Plane

 Right Plane

 Origin

▼  Revolve1

 Sketch1

▼  Cut-Extrude1

 Sketch2

 Chamfer1

▼  Cut-Extrude2

 Sketch3

Keyring Connector

Equations

3003 Alloy

Front Plane

Top Plane

Right Plane

Origin

Revolve1

Sketch1

Cut-Extrude1

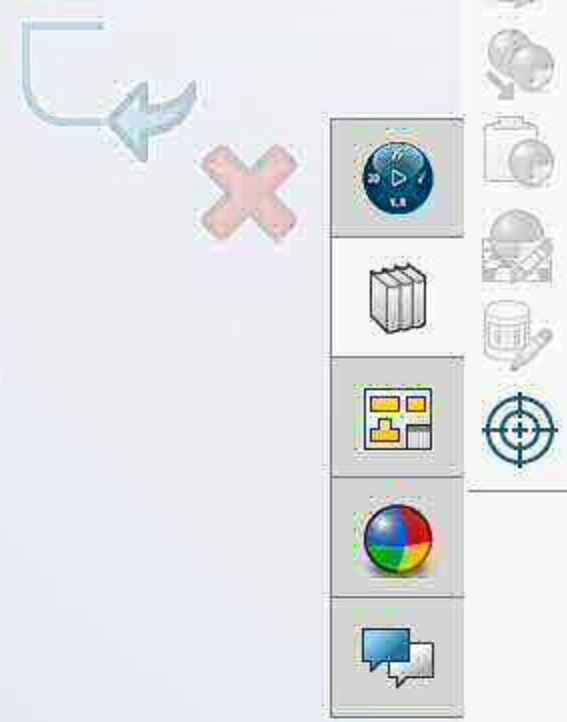
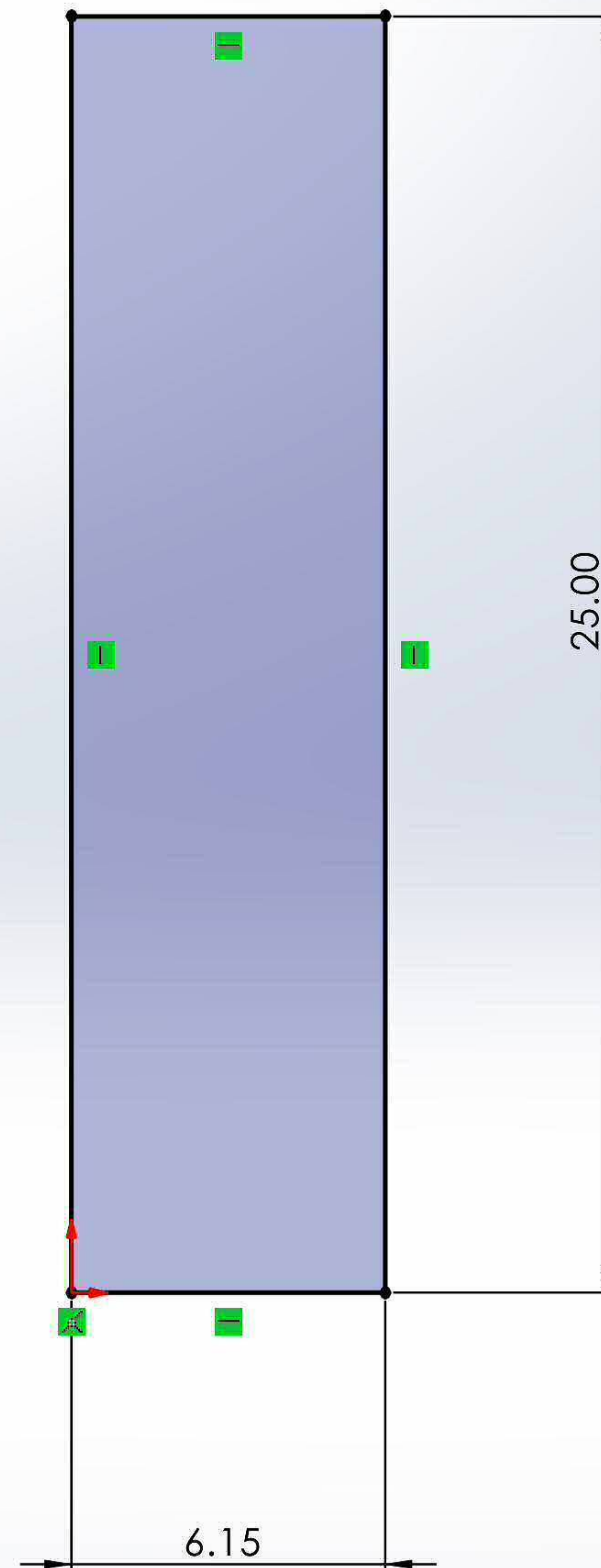
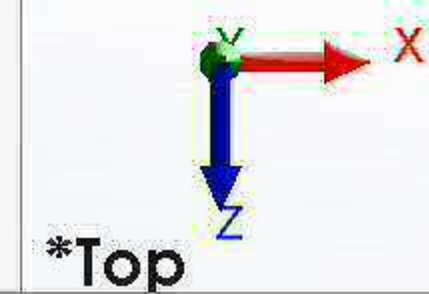
Sketch2

Chamfer1

Cut-Extrude2

Sketch3

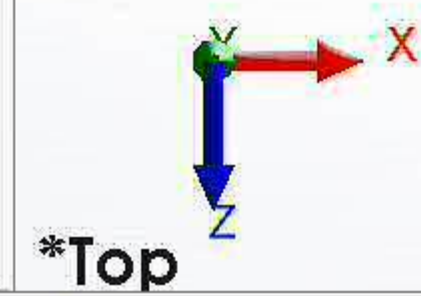
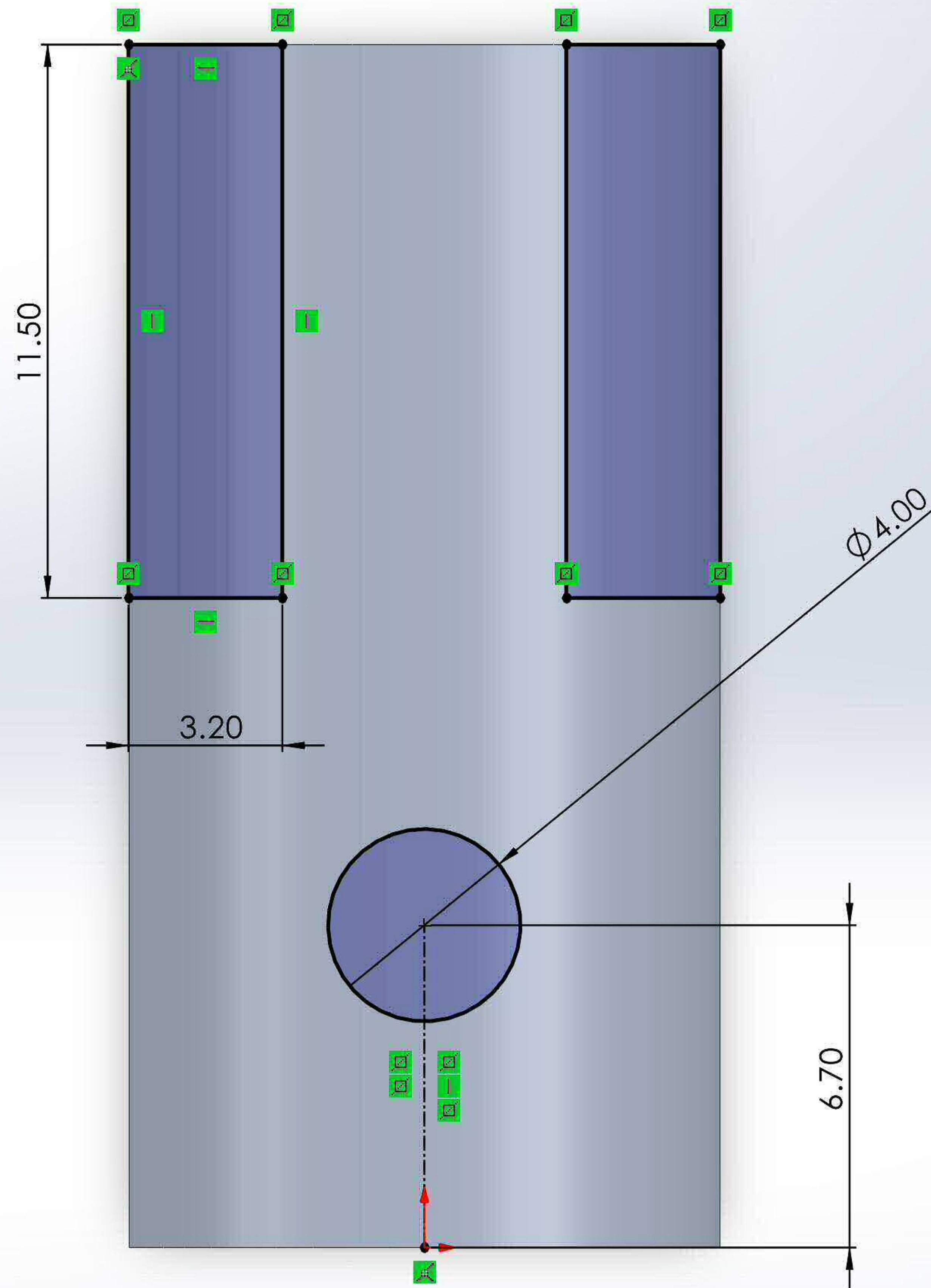
Sketch 1:



Keyring Connector

- Solid Bodies(1)
 - Revolve1
- Equations
- 3003 Alloy
- Front Plane
- Top Plane
- Right Plane
- Origin
- Revolve1
 - Sketch1
- Cut-Extrude1
 - Sketch2
- Chamfer1
- Cut-Extrude2
 - Sketch3

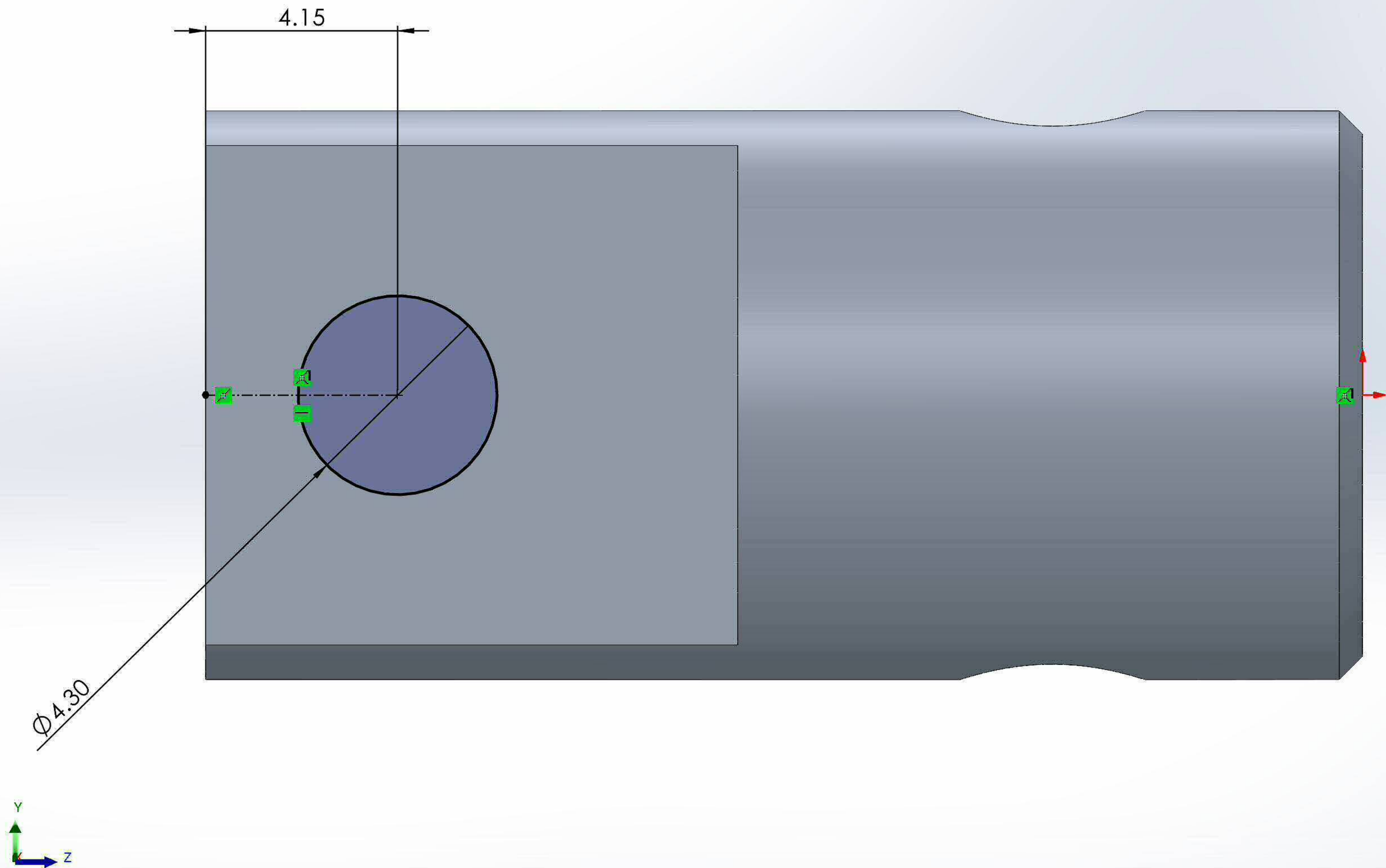
Sketch 2:



Keyring Connector

- Solid Bodies(1)
 - Chamfer1
- Equations
- 3003 Alloy
- Front Plane
- Top Plane
- Right Plane
- Origin
- Revolve1
 - Sketch1
- Cut-Extrude1
 - Sketch2
- Chamfer1
- Cut-Extrude2
 - Sketch3

Sketch 3:





Mass Properties:



Keyring Connector

Options...

Override Mass Properties...

Recalculate

☒ Include hidden bodies/components☐ Create Center of Mass feature☐ Show weld bead mass

Report coordinate values relative to: -- default --

Mass properties of Keyring Connector

Configuration: Default

Coordinate system: -- default --

Density = 0.00 grams per cubic millimeter

Mass = 5.84 grams

Volume = 2162.60 cubic millimeters

Surface area = 1313.00 square millimeters

Center of mass: (millimeters)

X = 0.00

Y = 0.00

Z = -10.84

Principal axes of inertia and principal moments of inertia: (grams * square millimeters)

Taken at the center of mass.

I_x = (0.00, 0.00, 1.00) P_x = 106.37I_y = (0.00, -1.00, 0.00) P_y = 328.45I_z = (1.00, 0.00, 0.00) P_z = 343.40

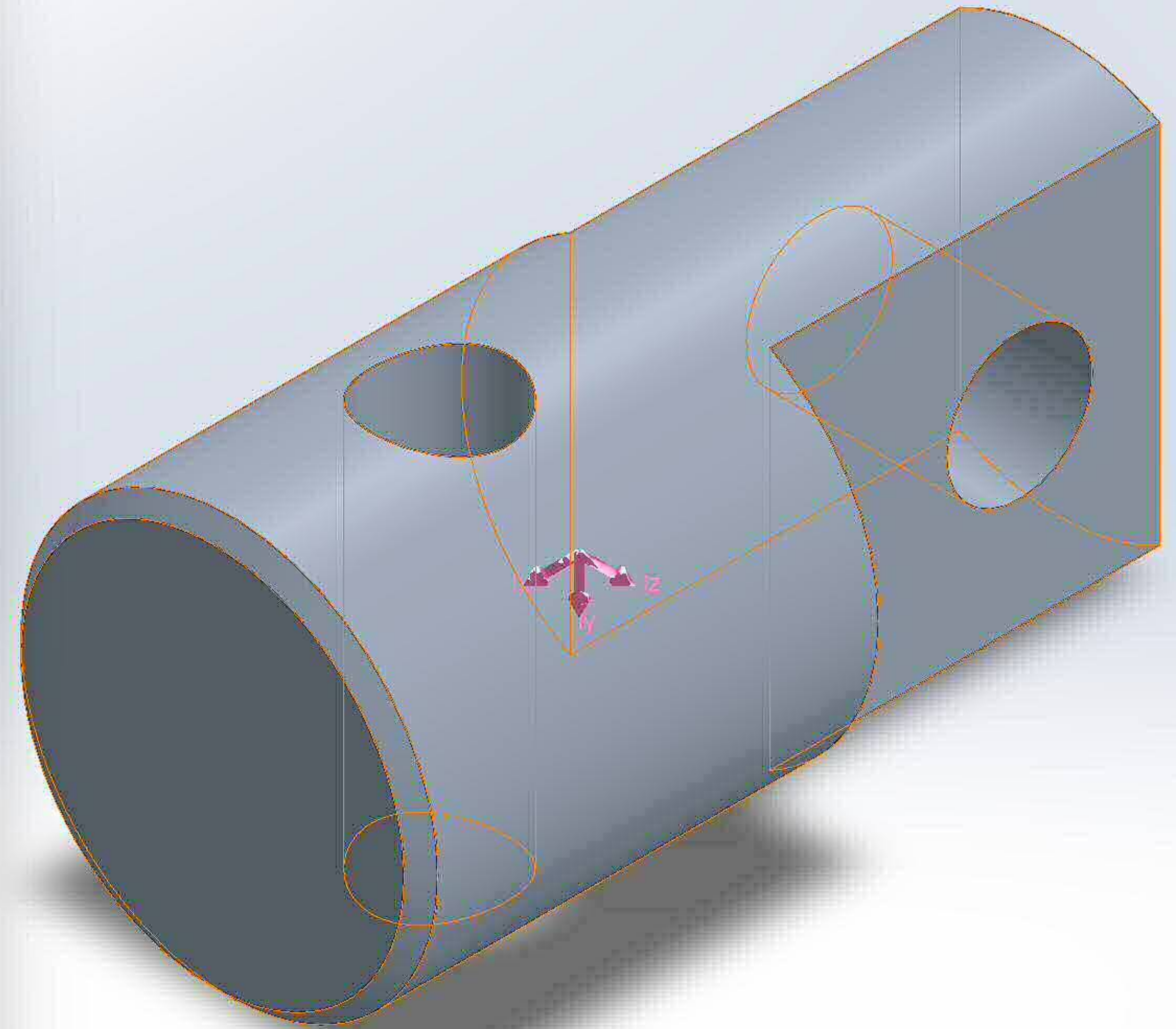
Moments of inertia: (grams * square millimeters)

Taken at the center of mass and aligned with the output coordinate system. (Using positive tensor notation.)

L_{xx} = 343.40 L_{xy} = 0.00 L_{xz} = 0.00L_{yx} = 0.00 L_{yy} = 328.45 L_{yz} = 0.00L_{zx} = 0.00 L_{zy} = 0.00 L_{zz} = 106.37

Moments of inertia: (grams * square millimeters)

Taken at the output coordinate system. (Using positive tensor notation.)

I_{xx} = 1029.65 I_{xy} = 0.00 I_{xz} = 0.00I_{yx} = 0.00 I_{yy} = 1014.71 I_{yz} = -0.01I_{zx} = 0.00 I_{zy} = -0.01 I_{zz} = 106.37

Units Used:

MKS (meter, kilogram, second)

CGS (centimeter, gram, second)

MMGS (millimeter, gram, second)

IPS (inch, pound, second)

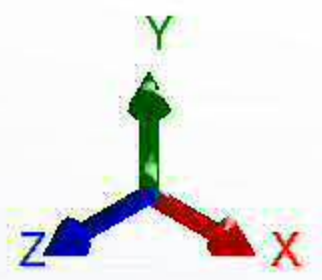
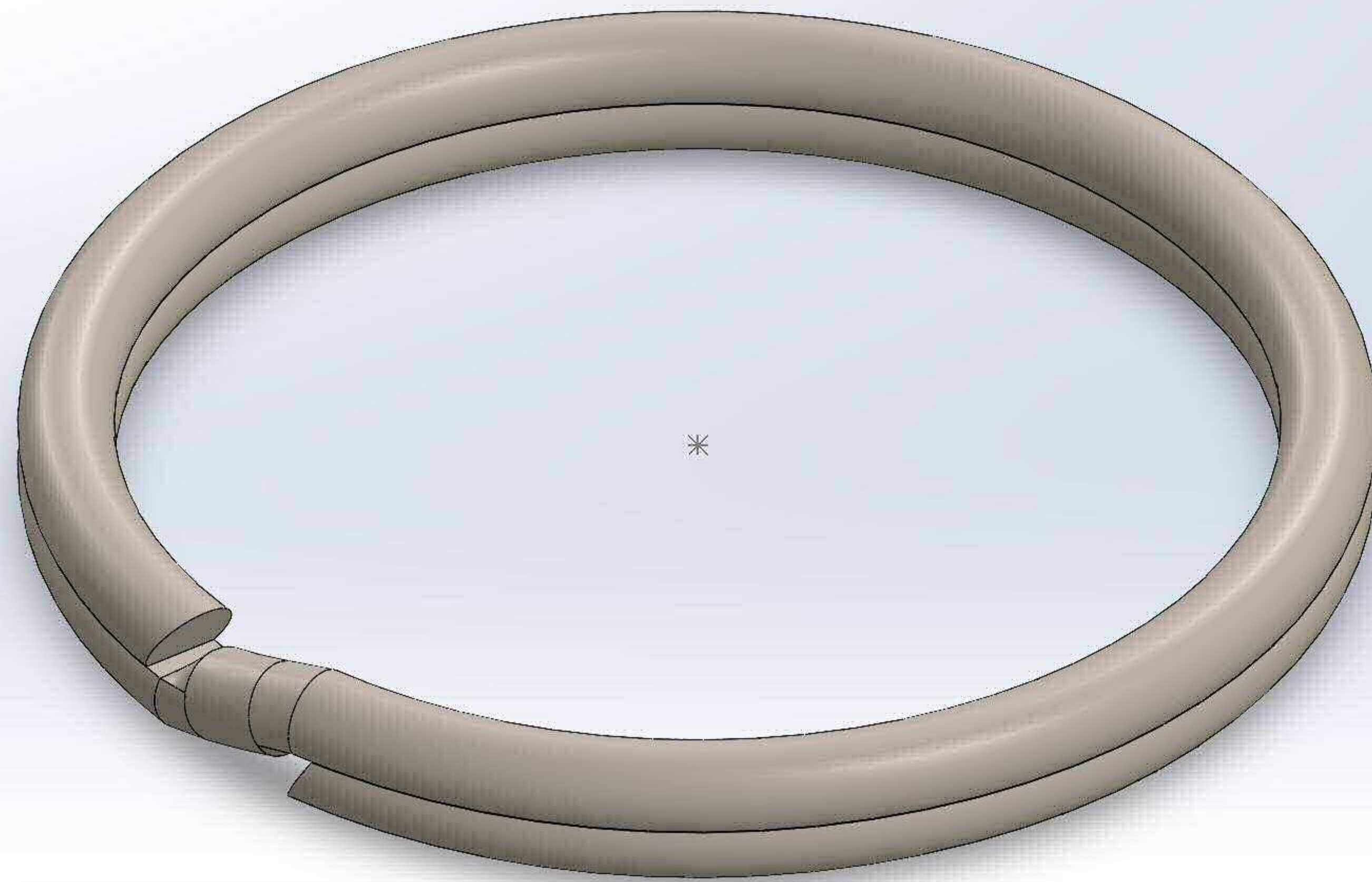
Edit Document Units...

Simplified Interface

MMGS



Isometric View:





Feature Tree:

-   Keyring
- ▼  Solid Bodies(1)
 -  TransitionToBottom
- ▼  Equations
 -  "InnerDiameter"=28.00mm
 -  "OuterDiameter"=32.64mm
 -  "WireDiameter"=2.32mm
-  Alloy Steel
-  Front Plane
-  Top Plane
-  Right Plane
-  Origin
- ▼  Primary Revolve
 -  Centre
 -  RevolveStart
 -  RevolveEndFlat
 -  RevolveEndAxis
 -  RevolveEndAngle
 - ▼  Revolve1
 -  Sketch1
 -  PatternAxis
 -  Patterned Ring
- ▼  Transition
 -  3D TransitionLine
 -  TransitionMidplane
 - ▼  TransitionRevolve
 -  Sketch2
 -  TransitionToTop
 -  TransitionToBottom
 -  Plane1
 -  Sketch3

Sketch 1:

Keyring

Equations

- "InnerDiameter"=28.00mm
- "OuterDiameter"=32.64mm
- "WireDiameter"=2.32mm

Alloy Steel

Front Plane

Top Plane

Right Plane

Origin

Primary Revolve

- Centre
- RevolveStart
- RevolveEndFlat
- RevolveEndAxis
- RevolveEndAngle

Revolve1

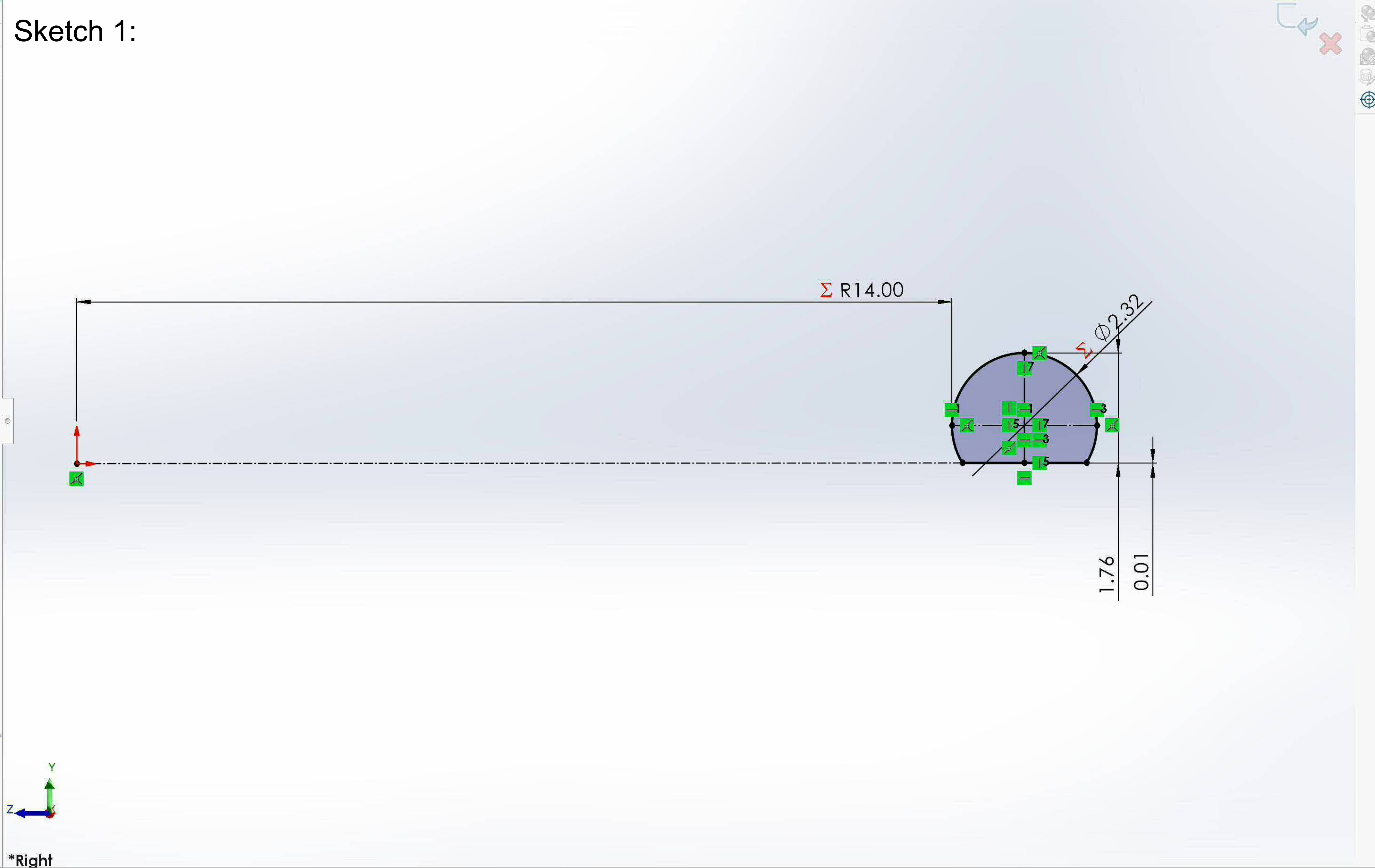
- Sketch1
- PatternAxis
- Patterned Ring

Transition

- TransitionLine
- TransitionMidplane
- TransitionRevolve
 - Sketch2
- TransitionToTop
- TransitionToBottom

Plane1

Sketch3



TransitionLine 3D Sketch:

Keyring

Solid Bodies(2)

- Revolve1
- Patterned Ring

Equations

- "InnerDiameter"=28.00mm
- "OuterDiameter"=32.64mm
- "WireDiameter"=2.32mm

Alloy Steel

Front Plane

Top Plane

Right Plane

Origin

Primary Revolve

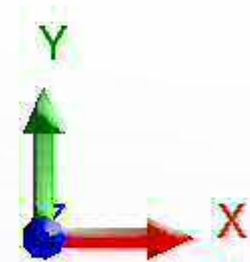
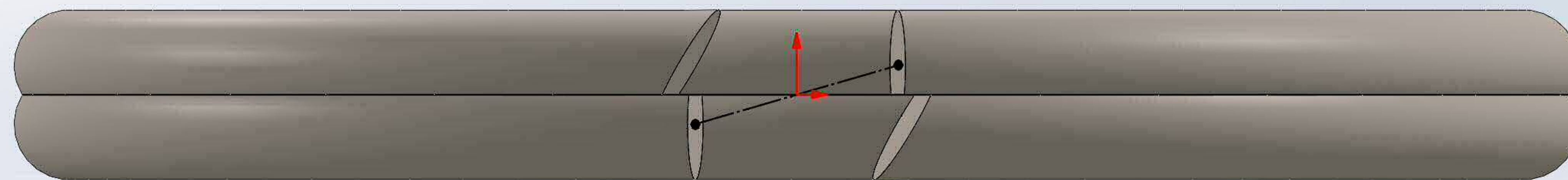
- Centre
- RevolveStart
- RevolveEndFlat
- RevolveEndAxis
- RevolveEndAngle
- Revolve1
 - Sketch1
- PatternAxis
- Patterned Ring

Transition

- 3D TransitionLine
- TransitionMidplane
- TransitionRevolve
 - Sketch2
- TransitionToTop
- TransitionToBottom

Plane1

Sketch3

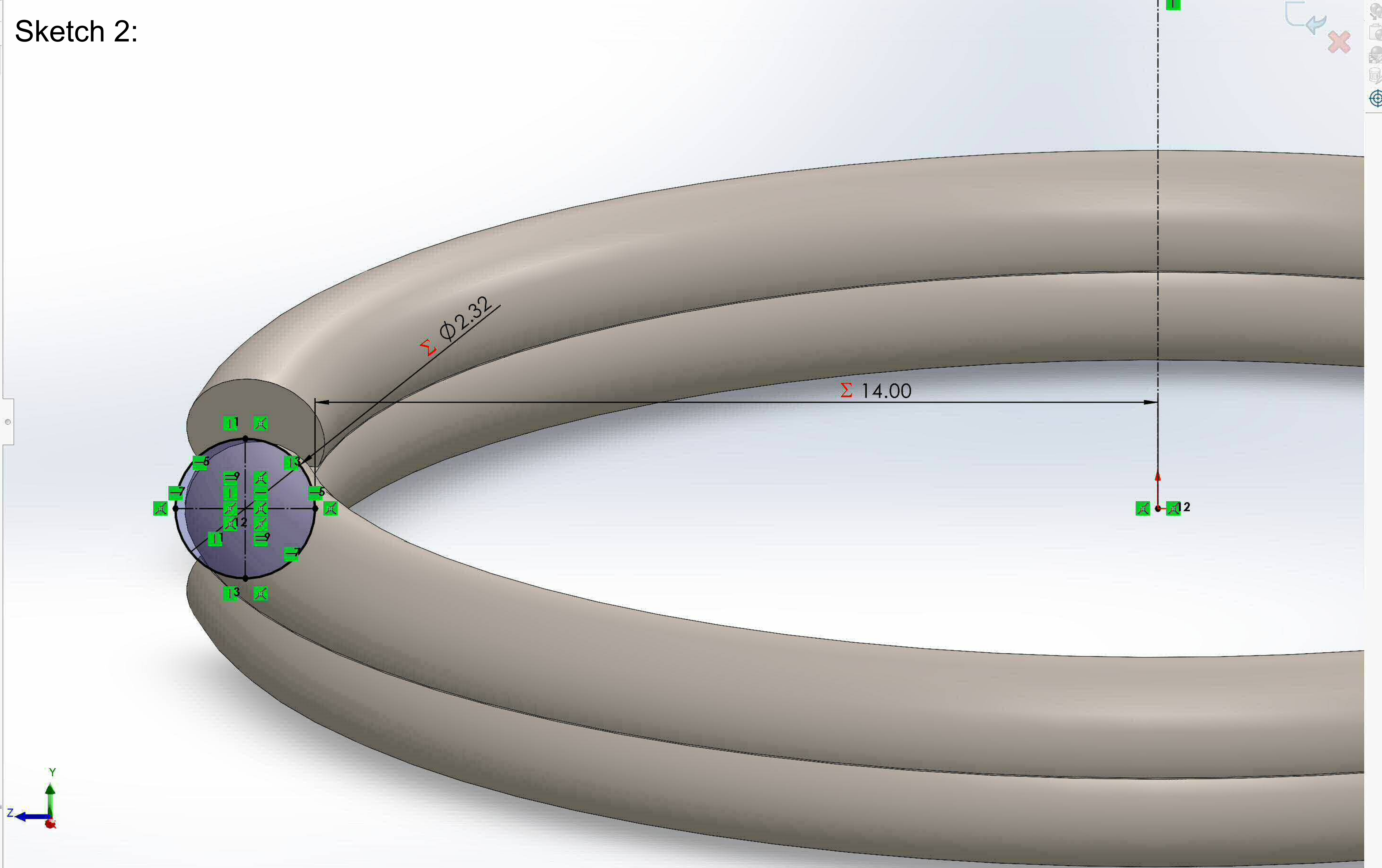


*Front

Sketch 2:

Keyring

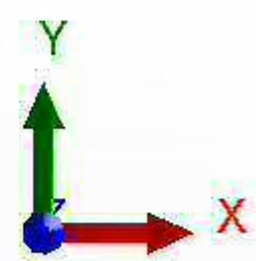
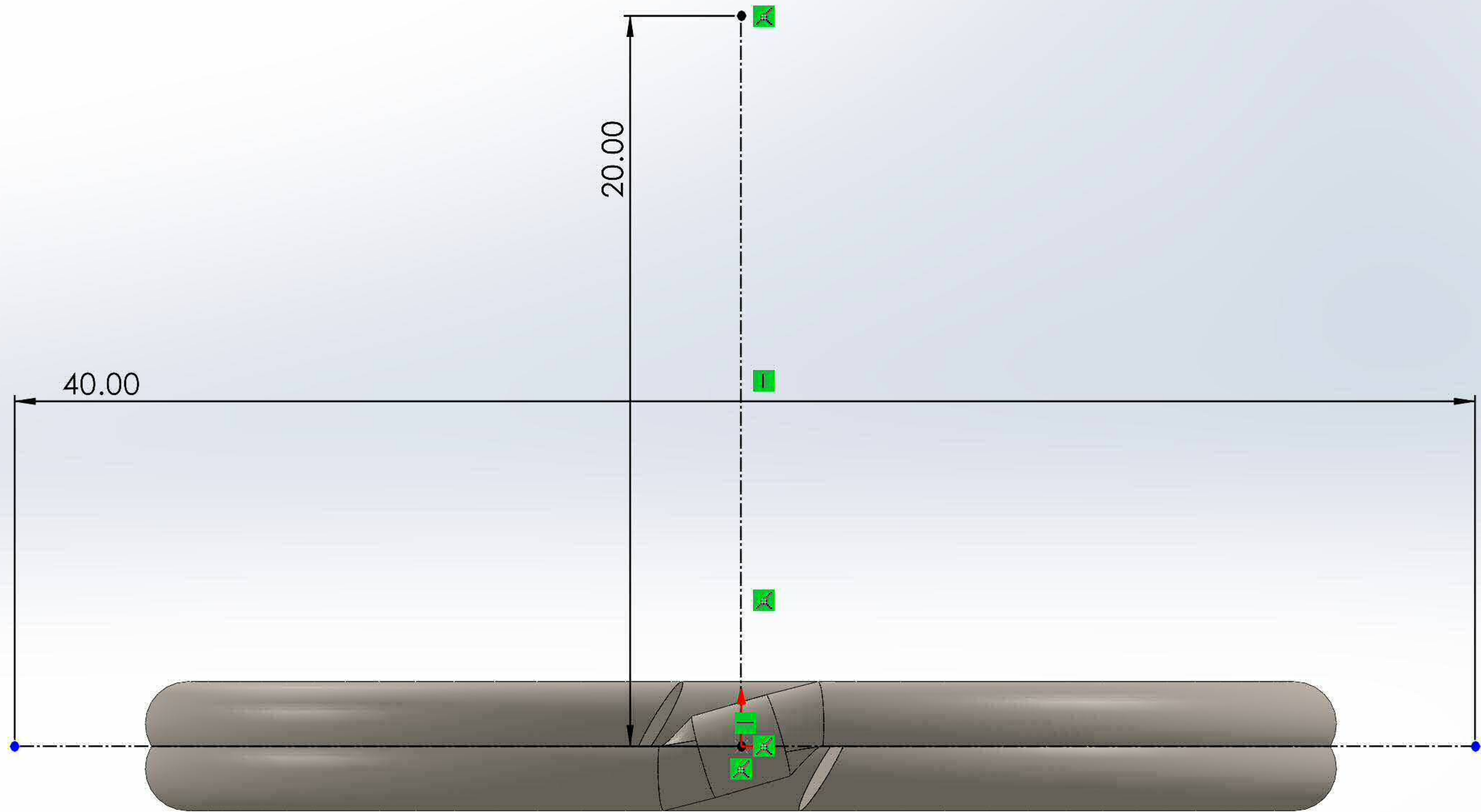
- Solid Bodies(2)
 - Revolve1
 - Patterned Ring
- Equations
 - "InnerDiameter" = 28.00mm
 - "OuterDiameter" = 32.64mm
 - "WireDiameter" = 2.32mm
- Alloy Steel
- Front Plane
- Top Plane
- Right Plane
- Origin
- Primary Revolve
 - Centre
 - RevolveStart
 - RevolveEndFlat
 - RevolveEndAxis
 - RevolveEndAngle
 - Revolve1
 - Sketch1
 - PatternAxis
 - Patterned Ring
- Transition
 - 3D TransitionLine
 - TransitionMidplane
 - TransitionRevolve
 - Sketch2
 - TransitionToTop
 - TransitionToBottom
- Plane1
- Sketch3



Sketch 3:

Keyring

- Solid Bodies(1)
 - TransitionToBottom
- Equations
 - "InnerDiameter"=28.00mm
 - "OuterDiameter"=32.64mm
 - "WireDiameter"=2.32mm
- Alloy Steel
- Front Plane
- Top Plane
- Right Plane
- Origin
- Primary Revolve
 - Centre
 - RevolveStart
 - RevolveEndFlat
 - RevolveEndAxis
 - RevolveEndAngle
- Revolve1
 - Sketch1
 - PatternAxis
 - Patterned Ring
- Transition
 - TransitionLine
 - TransitionMidplane
 - TransitionRevolve
 - Sketch2
 - TransitionToTop
 - TransitionToBottom
- Plane1
- Sketch3





Mass Properties:



Keyring

Options...

Override Mass Properties...

Recalculate

☒ Include hidden bodies/components☐ Create Center of Mass feature☐ Show weld bead mass

Report coordinate values relative to: -- default --

Mass properties of Keyring

Configuration: Default

Coordinate system: -- default --

Density = 0.01 grams per cubic millimeter

Mass = 4.96 grams

Volume = 643.62 cubic millimeters

Surface area = 1293.49 square millimeters

Center of mass: (millimeters)

X = 0.00

Y = 0.00

Z = -0.28

Principal axes of inertia and principal moments of inertia: (grams * square millimeters)

Taken at the center of mass.

I_x = (1.00, 0.00, 0.00) Px = 565.57I_y = (0.00, 0.00, -1.00) Py = 586.79I_z = (0.00, 1.00, 0.00) Pz = 1143.97

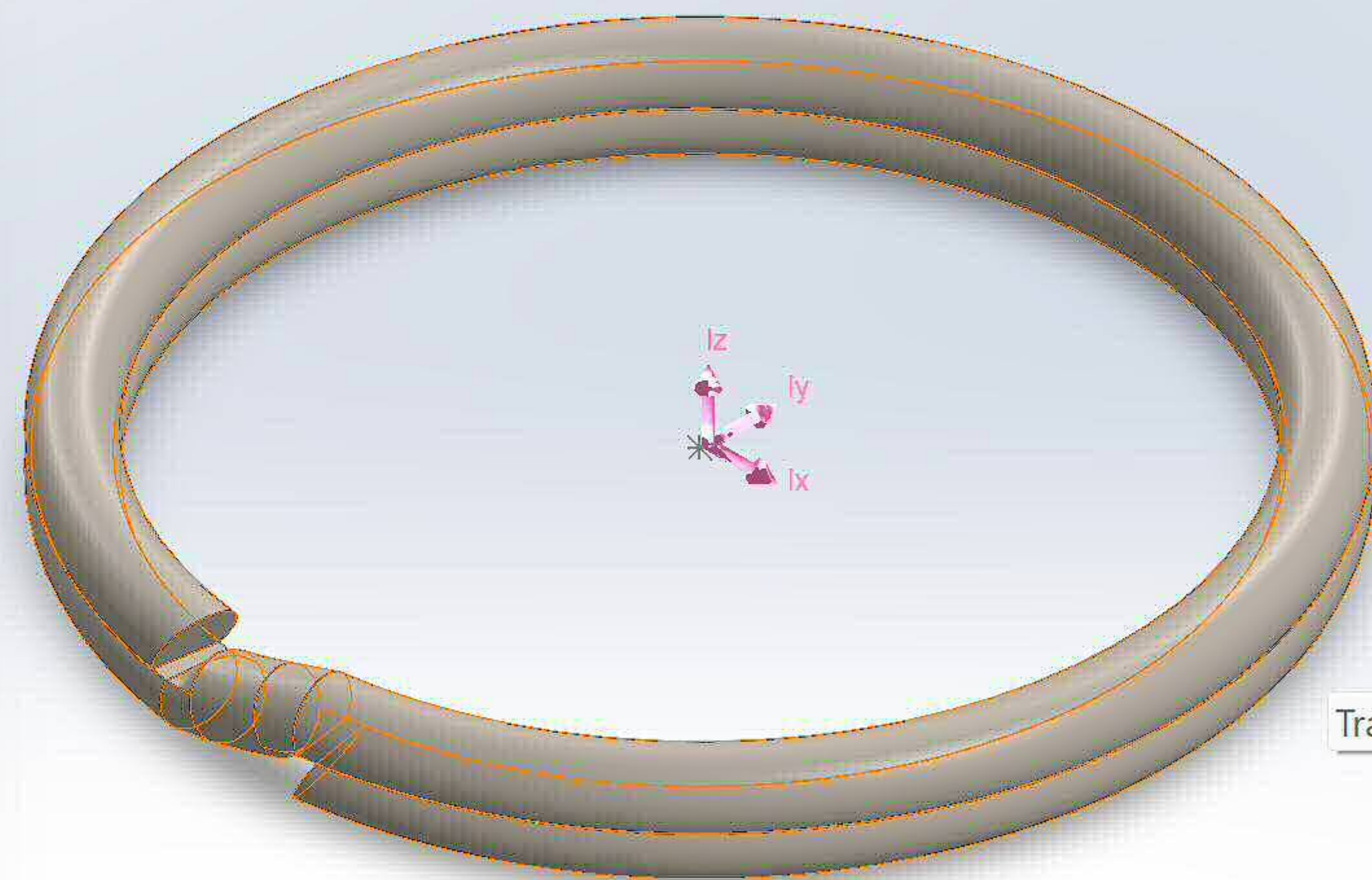
Moments of inertia: (grams * square millimeters)

Taken at the center of mass and aligned with the output coordinate system. (Using positive tensor notation.)

L_{xx} = 565.57 L_{xy} = 0.06 L_{xz} = 0.00L_{yx} = 0.06 L_{yy} = 1143.97 L_{yz} = 0.00L_{zx} = 0.00 L_{zy} = 0.00 L_{zz} = 586.79

Moments of inertia: (grams * square millimeters)

Taken at the output coordinate system. (Using positive tensor notation.)

I_{xx} = 565.95 I_{xy} = 0.06 I_{xz} = 0.00I_{yx} = 0.06 I_{yy} = 1144.36 I_{yz} = 0.00I_{zx} = 0.00 I_{zy} = 0.00 I_{zz} = 586.79

Trans.

Units Used:

MKS (meter, kilogram, second)

CGS (centimeter, gram, second)

MMGS (millimeter, gram, second)

IPS (inch, pound, second)

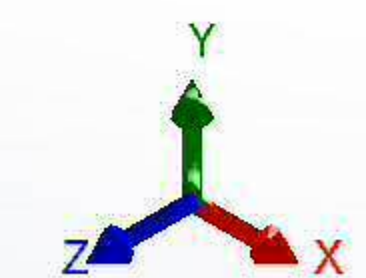
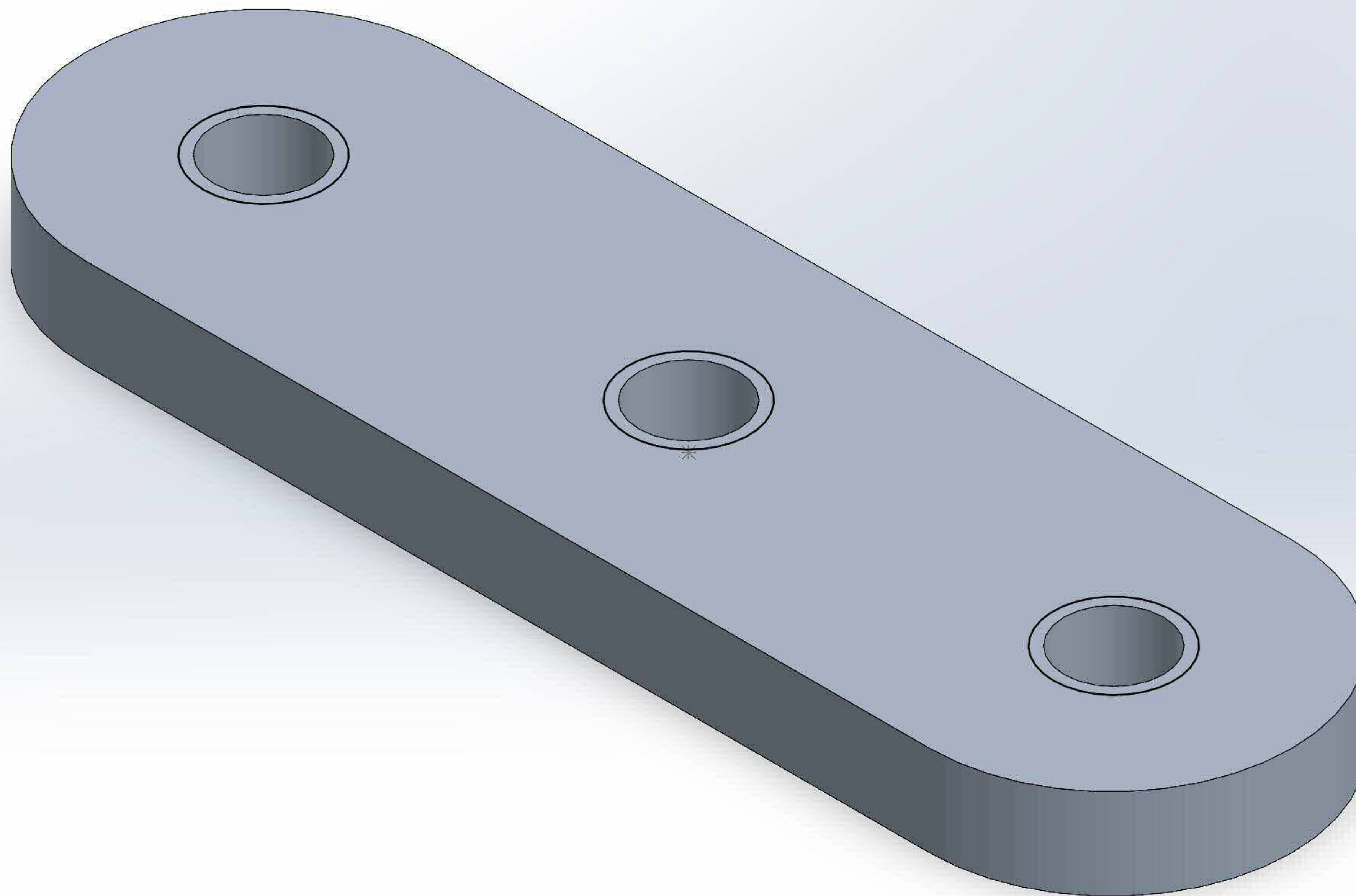
Edit Document Units...

Simplified Interface

MMGS



Isometric View:





Feature Tree:

Bottom Plate

- ▼  Solid Bodies(1)
 -  M4x0.7 Tapped Hole1
-  Equations
-  3003 Alloy
-  Front Plane
-  Top Plane
-  Right Plane
-  Origin
- ▼  Boss-Extrude1
 -  Sketch1
- ▼  M4x0.7 Tapped Hole1
 -  Sketch2
 -  Sketch3
 -  Hole Thread1
 -  Hole Thread2
 -  Hole Thread3

Sketch 1:

Bottom Plate

Equations

3003 Alloy

Front Plane

Top Plane

Right Plane

Origin

Boss-Extrude1

Sketch1

M4x0.7 Tapped Hole1

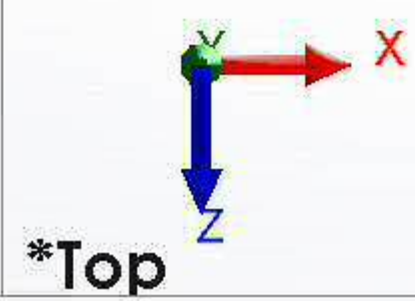
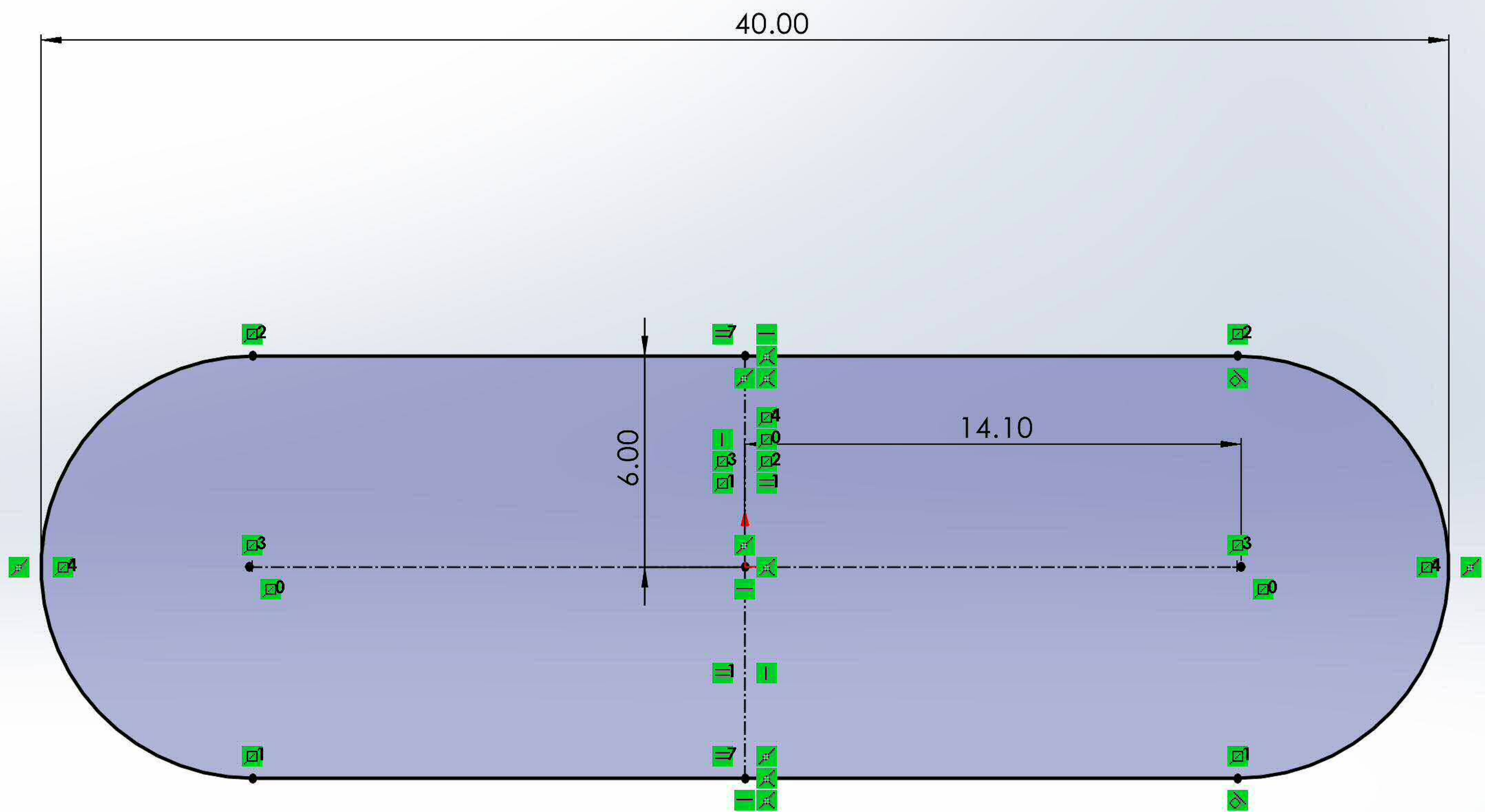
Sketch2

Sketch3

Hole Thread1

Hole Thread2

Hole Thread3





Hole Specification

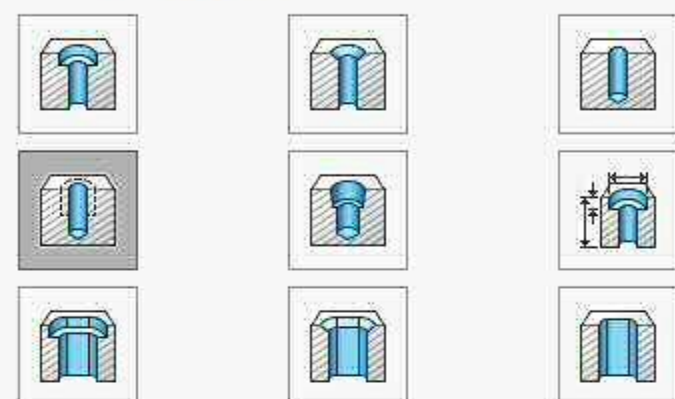


Hole Wizard:

 Type  Positions

No Favorite Selected

Hole Type



Standard:

ANSI Metric

Type:

Bottoming Tapped Hole

Hole Specifications

Size:

M4x0.7



Show custom sizing

End Condition



Through All

Thread:

Through All

Options



☒ With thread callout

☐ Thread class

☐ Near side countersink

☐ Far side countersink

Tolerance/Precision

Callout value

Thru Tap Drill Diameter

+0.01
1.50
-0.01

None

.01
X.XXX
.01

.12 (Document)

Sketch 3: (Created By The Hole Wizard)

Bottom Plate

Solid Bodies(1)

Boss-Extrude1

Equations

3003 Alloy

Front Plane

Top Plane

Right Plane

Origin

Boss-Extrude1

Sketch1

M4x0.7 Tapped Hole1

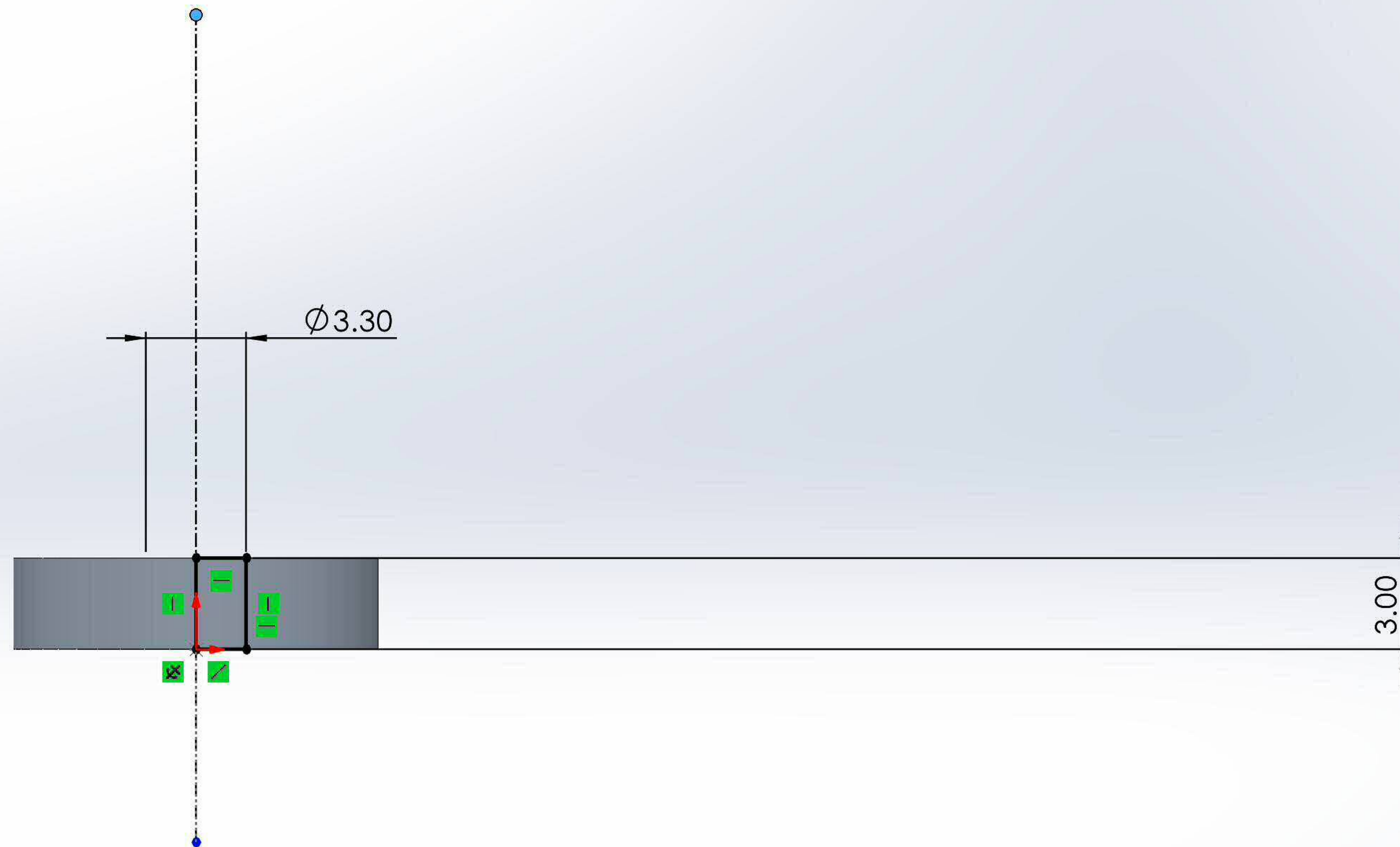
Sketch2

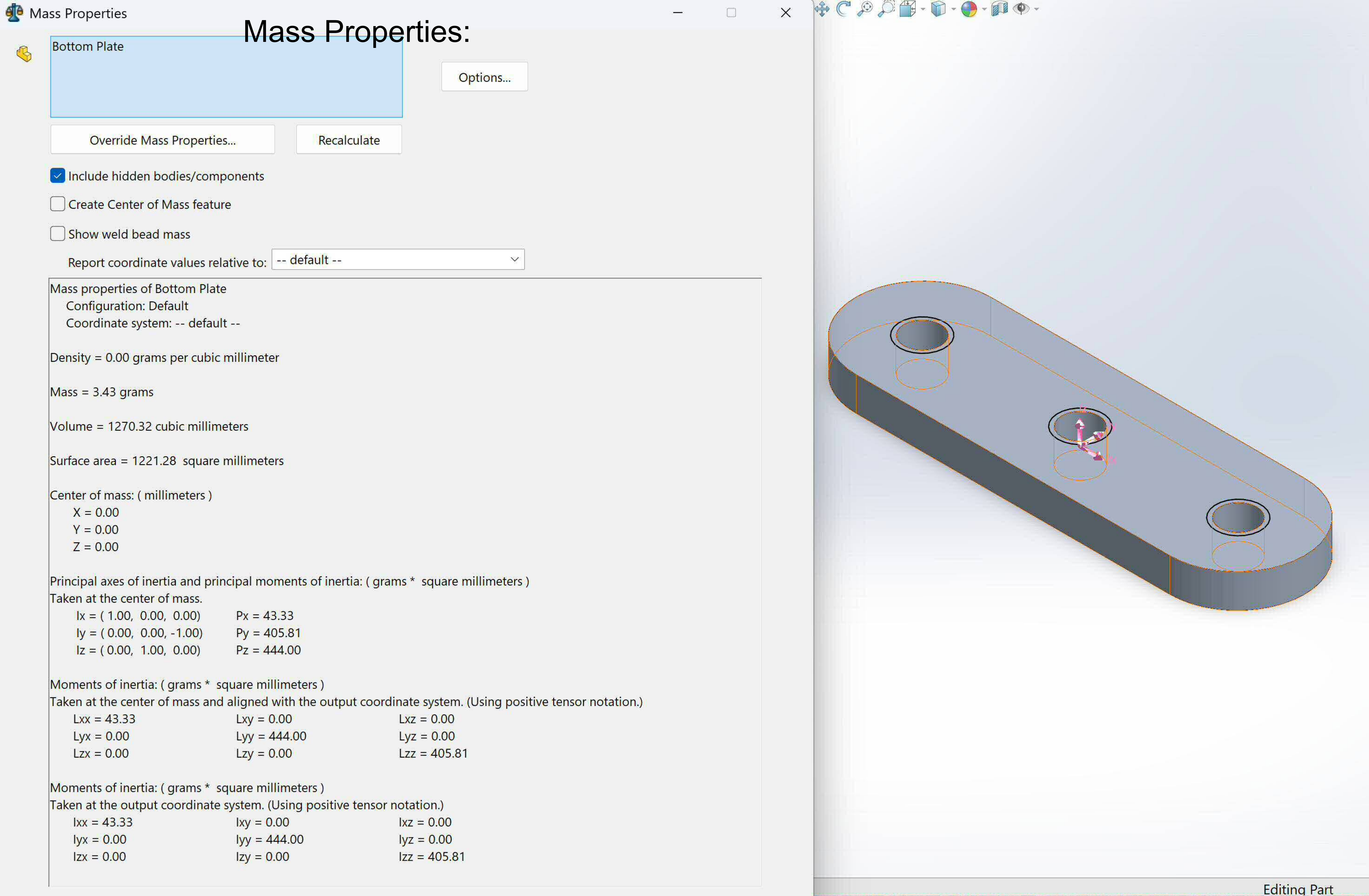
Sketch3

Hole Thread1

Hole Thread2

Hole Thread3





Mass Properties:

Bottom Plate

Options...

Override Mass Properties...

Recalculate

☒ Include hidden bodies/components

☐ Create Center of Mass feature

☐ Show weld bead mass

Report coordinate values relative to: -- default --

Mass properties of Bottom Plate

Configuration: Default

Coordinate system: -- default --

Density = 0.00 grams per cubic millimeter

Mass = 3.43 grams

Volume = 1270.32 cubic millimeters

Surface area = 1221.28 square millimeters

Center of mass: (millimeters)

X = 0.00

Y = 0.00

Z = 0.00

Principal axes of inertia and principal moments of inertia: (grams * square millimeters)

Taken at the center of mass.

Ix = (1.00, 0.00, 0.00) Px = 43.33

Iy = (0.00, 0.00, -1.00) Py = 405.81

Iz = (0.00, 1.00, 0.00) Pz = 444.00

Moments of inertia: (grams * square millimeters)

Taken at the center of mass and aligned with the output coordinate system. (Using positive tensor notation.)

Lxx = 43.33 Lxy = 0.00 Lxz = 0.00

Lyx = 0.00 Lyy = 444.00 Lyz = 0.00

Lzx = 0.00 Lzy = 0.00 Lzz = 405.81

Moments of inertia: (grams * square millimeters)

Taken at the output coordinate system. (Using positive tensor notation.)

Ixx = 43.33 Ixy = 0.00 Ixz = 0.00

Iyx = 0.00 Iyy = 444.00 Iyz = 0.00

Izx = 0.00 Izy = 0.00 Izz = 405.81

Units Used:

MKS (meter, kilogram, second)

CGS (centimeter, gram, second)

MMGS (millimeter, gram, second)

IPS (inch, pound, second)

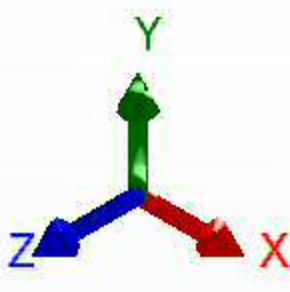
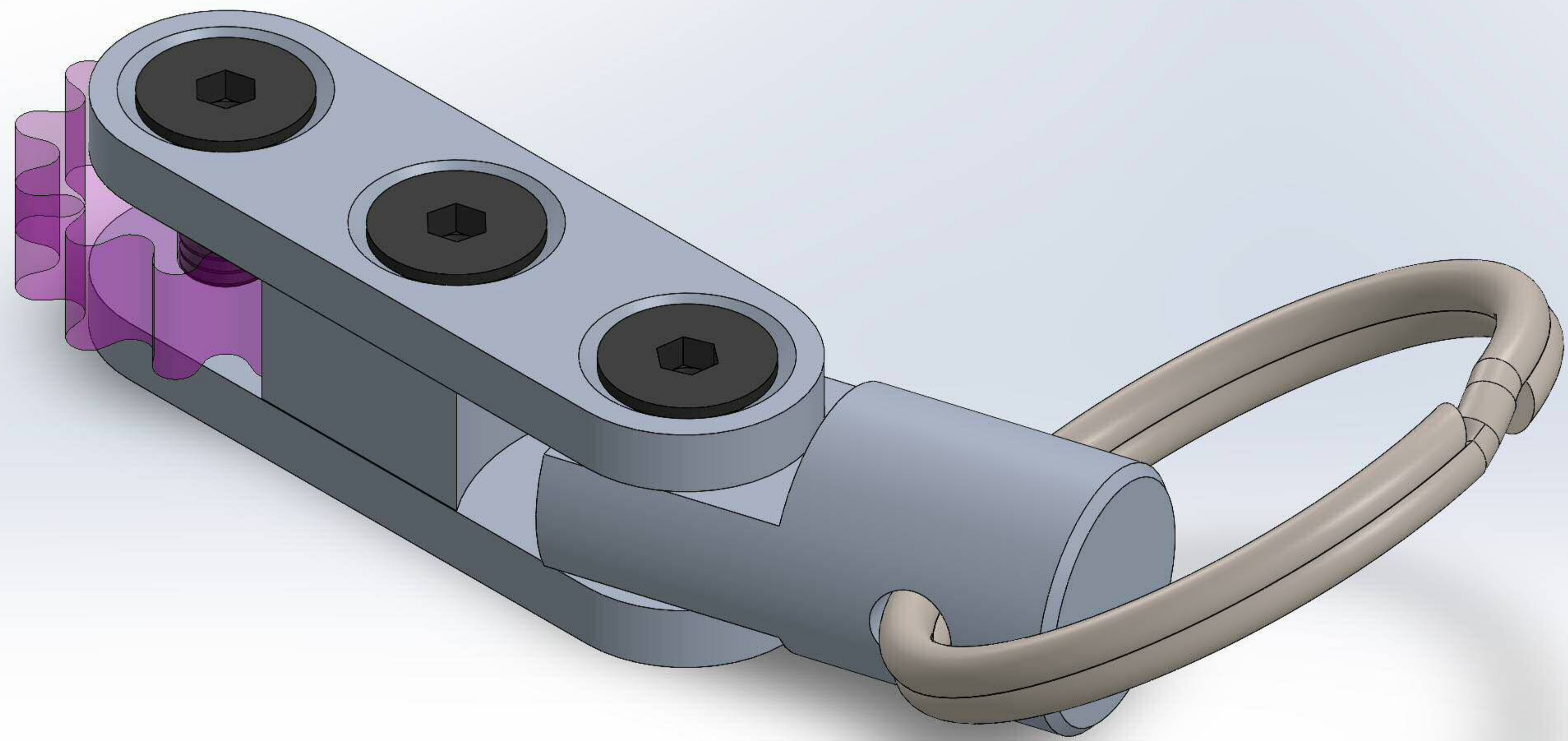
Edit Document Units...

Simplified Interface

MMGS

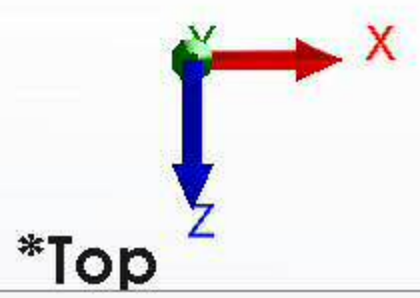
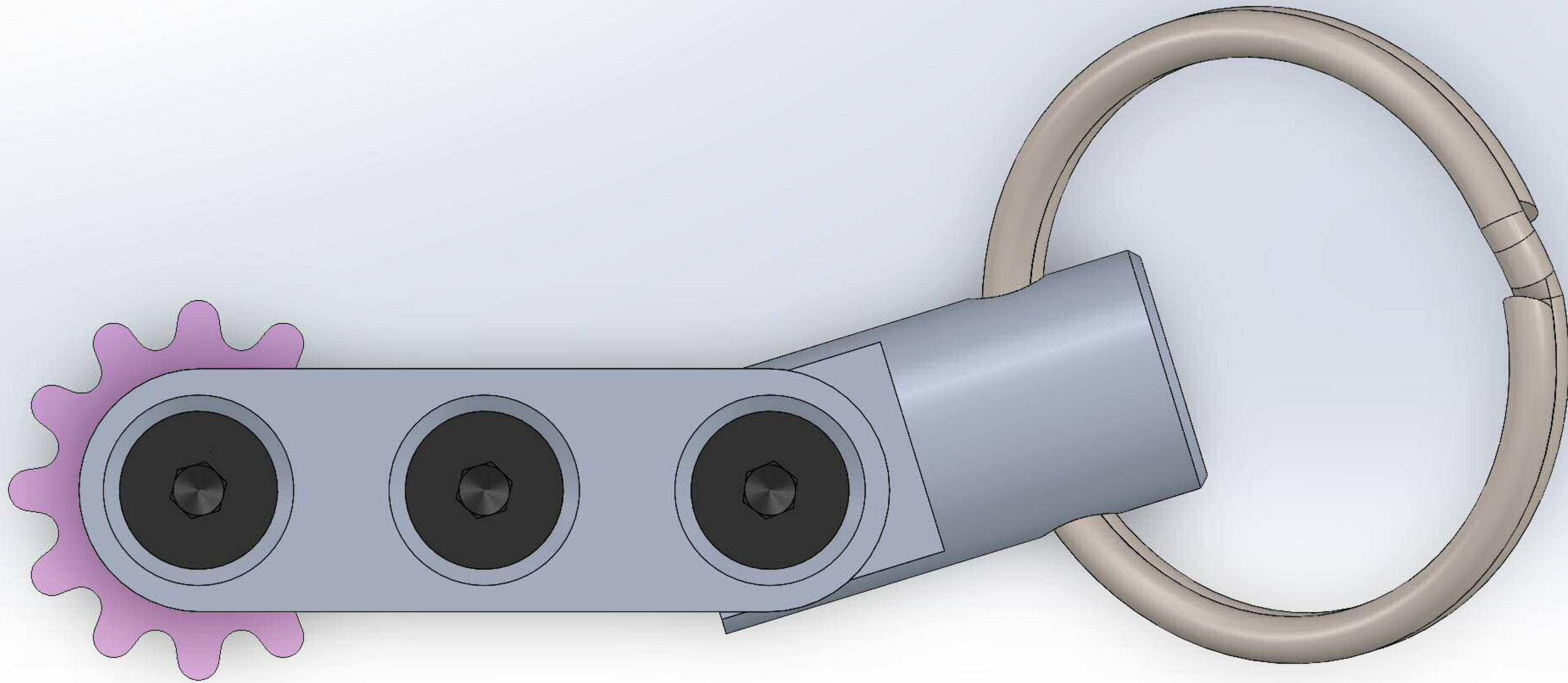


Isometric View:

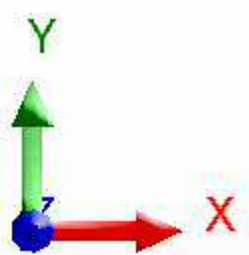
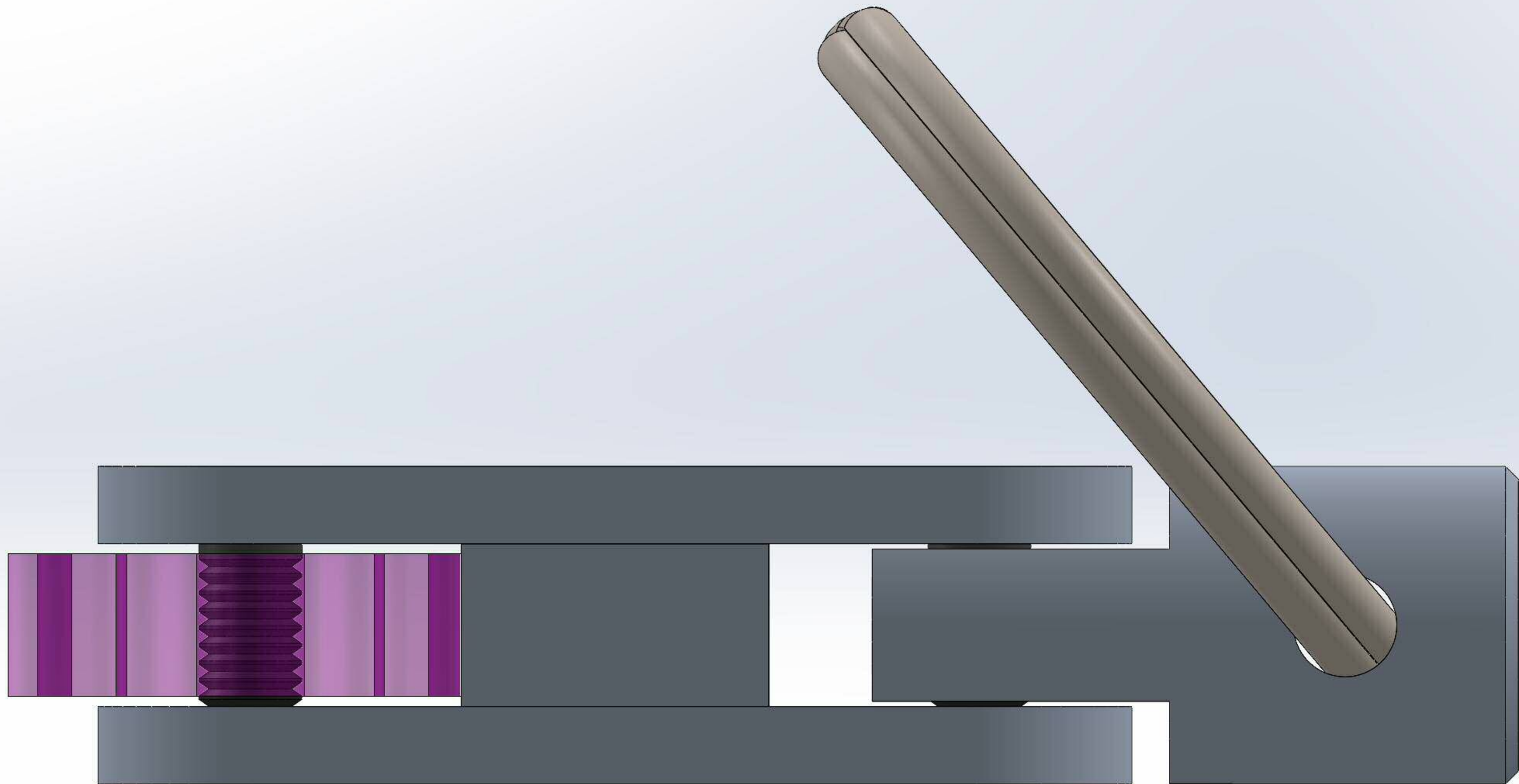


*Isometric

Top View:



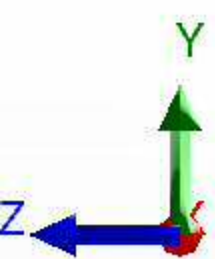
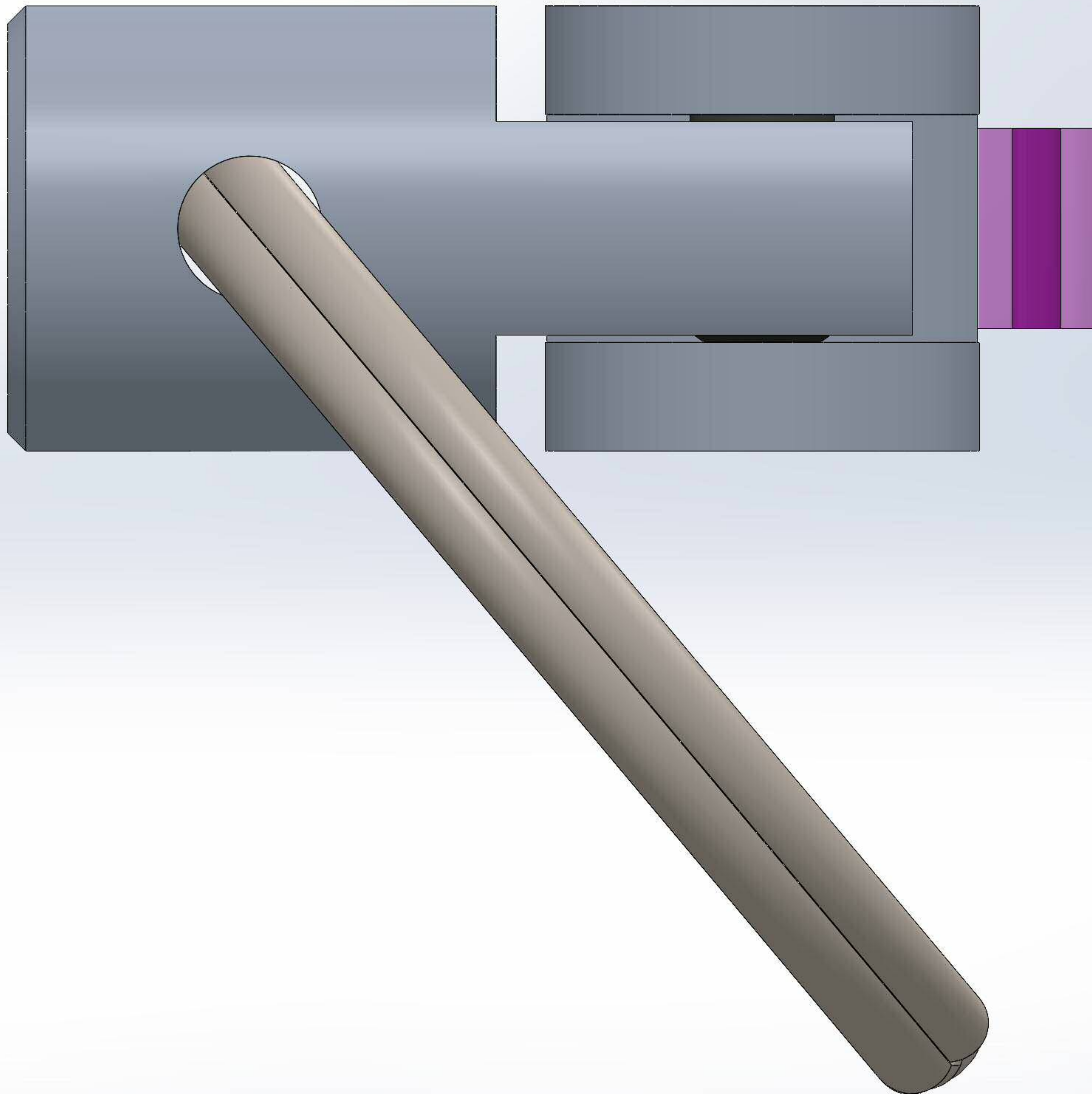
Front View:



*Front

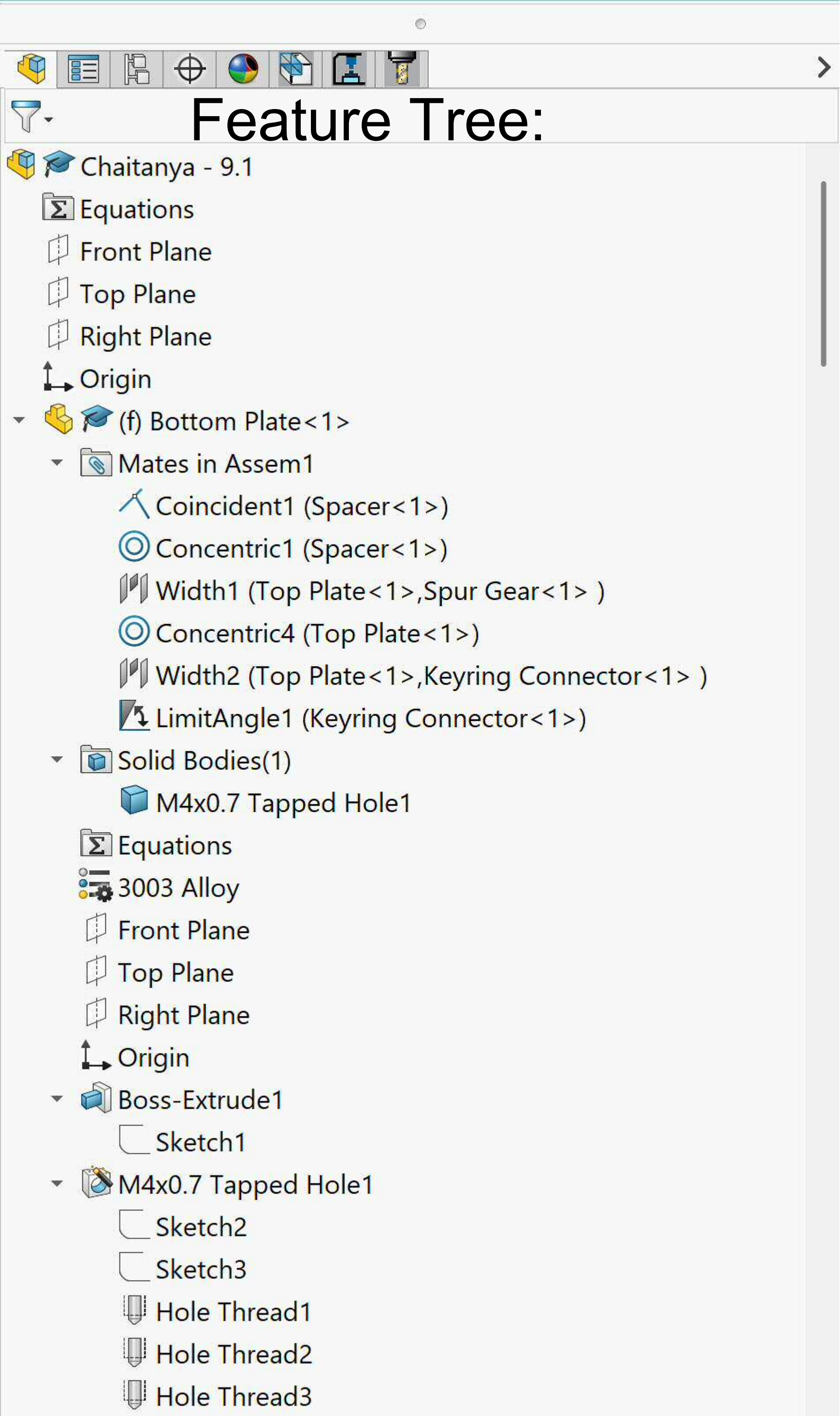


Right View:



*Right





▼   (-) Spacer<1>

▼  Mates in Assem1

  Coincident1 (Bottom Plate<1>)

  Concentric1 (Bottom Plate<1>)

  Coincident2 (Top Plate<1>)

  Concentric2 (Top Plate<1>)

 Concentric7 (M4 Screw<2>)

▼  Solid Bodies(1)

 Boss-Extrude1

 Equations

 3003 Alloy

 Front Plane

 Top Plane

 Right Plane

 Origin

▼  Boss-Extrude1

 Sketch1

▼ Top Plate<1>

▼ Mates in Assem1

  Coincident2 (Spacer<1>)

  Concentric2 (Spacer<1>)

  Width1 (Bottom Plate<1>,Spur Gear<1>)

  Concentric3 (Spur Gear<1>)

  Concentric4 (Bottom Plate<1>)

  Concentric5 (Keyring Connector<1>)

  Width2 (Bottom Plate<1>,Keyring Connector<1>)

 Coincident3 (M4 Screw<1>)

 Coincident4 (M4 Screw<2>)

 Coincident5 (M4 Screw<3>)

 LimitAngle2 (Keyring<1>)

▼ Solid Bodies(1)

 CSK for M4 Flat Head Machine Screw1

 Equations

 3003 Alloy

 Front Plane

 Top Plane

 Right Plane

 Origin

▼ Boss-Extrude1

 Sketch2

▼ CSK for M4 Flat Head Machine Screw1

 Sketch3

 Sketch4

▼   (-) Spur Gear<1>

▼  Mates in Assem1

  Width1 (Bottom Plate<1>,Top Plate<1>)

  Concentric3 (Top Plate<1>)

 Concentric6 (M4 Screw<1>)

▼  Solid Bodies(1)

 Cut-Extrude1

 Equations

 Acrylic (Medium-high impact)

 Front Plane

 Top Plane

 Right Plane

 Origin

▼  Boss-Extrude1

 Sketch2

▼  Cut-Extrude1

 Sketch3

- ▼  Keyring Connector<1>
 - ▼  Mates in Assem1
 -  Concentric5 (Top Plate<1>)
 -  Width2 (Bottom Plate<1>,Top Plate<1>)
 -  LimitAngle1 (Bottom Plate<1>)
 -  Concentric8 (M4 Screw<3>)
 -  Coincident9 (Keyring<1>)
 -  Coincident10 (Keyring<1>)
 - ▼  Solid Bodies(1)
 -  Cut-Extrude2
 -  Equations
 -  3003 Alloy
 -  Front Plane
 -  Top Plane
 -  Right Plane
 -  Origin
 - ▼  Revolve1
 -  Sketch1
 - ▼  Cut-Extrude1
 -  Sketch2
 -  Chamfer1
 - ▼  Cut-Extrude2
 -  Sketch3

- ▼   (-) M4 Screw<1>
- ▼  Mates in Assem1
 -   Concentric6 (Spur Gear<1>)
 -   Coincident3 (Top Plate<1>)
- ▼  Solid Bodies(1)
 -  ThdSchPat
-  Equations
-  AISI 1020
-  Plane1
-  Plane2
-  Plane3
-  Origin
- ▼  MateReferences
 -  MateReference
- ▼  BaseBody
 -  BodySke
 -  ThreadCosmetic
-  Fillet1
- ▼  PreBroach
 -  Sketch2
- ▼  Hex
 -  Sketch3
- ▼  ThreadSchematic
 -  ThdSchSke
-  ThdSchPat

- ▼   (-) M4 Screw<2>
- ▼  Mates in Assem1
 -   Concentric7 (Spacer<1>)
 -   Coincident4 (Top Plate<1>)
- ▼  Solid Bodies(1)
 -  ThdSchPat
 -  Equations
 -  AISI 1020
 -  Plane1
 -  Plane2
 -  Plane3
 -  Origin
- ▼  MateReferences
 -  MateReference
- ▼  BaseBody
 -  BodySke
 -  ThreadCosmetic
 -  Fillet1
- ▼  PreBroach
 -  Sketch2
- ▼  Hex
 -  Sketch3
- ▼  ThreadSchematic
 -  ThdSchSke
 -  ThdSchPat

- ▼   (-) M4 Screw<3>
- ▼  Mates in Assem1
 -   Concentric8 (Keyring Connector<1>)
 -   Coincident5 (Top Plate<1>)
- ▼  Solid Bodies(1)
 -  ThdSchPat
-  Equations
-  AISI 1020
-  Plane1
-  Plane2
-  Plane3
-  Origin
- ▼  MateReferences
 -  MateReference
- ▼  BaseBody
 -  BodySke
 -  ThreadCosmetic
-  Fillet1
- ▼  PreBroach
 -  Sketch2
- ▼  Hex
 -  Sketch3
- ▼  ThreadSchematic
 -  ThdSchSke
-  ThdSchPat

- ▼   Keyring<1>
 - ▼  Mates in Assem1
 -   Coincident9 (Keyring Connector<1>)
 -   Coincident10 (Keyring Connector<1>)
 -   LimitAngle2 (Top Plate<1>)
 - ▼  Solid Bodies(1)
 -  TransitionToBottom
 - ▼  Equations
 -  "InnerDiameter"=28.00mm
 -  "OuterDiameter"=32.64mm
 -  "WireDiameter"=2.32mm
 -  Alloy Steel
 -  Front Plane
 -  Top Plane
 -  Right Plane
 -  Origin
 - ▼  Primary Revolve
 -  Centre
 -  RevolveStart
 -  RevolveEndFlat
 -  RevolveEndAxis
 -  RevolveEndAngle
 - ▼  Revolve1
 -  Sketch1
 -  PatternAxis
 -  Patterned Ring
 - ▼  Transition
 -  3D TransitionLine
 -  TransitionMidplane
 - ▼  TransitionRevolve
 -  Sketch2
 -  TransitionToTop
 -  TransitionToBottom
 -  Plane1
 -  Sketch3

▼ Mates

-  Coincident1 (Bottom Plate<1>,Spacer<1>)
-  Concentric1 (Bottom Plate<1>,Spacer<1>)
-  Coincident2 (Spacer<1>,Top Plate<1>)
-  Concentric2 (Spacer<1>,Top Plate<1>)
-  Width1 (Bottom Plate<1>,Top Plate<1>,Spur Gear<1>)
-  Concentric3 (Top Plate<1>,Spur Gear<1>)
-  Concentric4 (Bottom Plate<1>,Top Plate<1>)
-  Concentric5 (Top Plate<1>,Keyring Connector<1>)
-  Width2 (Bottom Plate<1>,Top Plate<1>,Keyring Connector<1>)
-  LimitAngle1 (Bottom Plate<1>,Keyring Connector<1>)
-  Concentric6 (Spur Gear<1>,M4 Screw<1>)
-  Coincident3 (Top Plate<1>,M4 Screw<1>)
-  Concentric7 (Spacer<1>,M4 Screw<2>)
-  Coincident4 (Top Plate<1>,M4 Screw<2>)
-  Concentric8 (Keyring Connector<1>,M4 Screw<3>)
-  Coincident5 (Top Plate<1>,M4 Screw<3>)
-  Coincident9 (Keyring Connector<1>,Keyring<1>)
-  Coincident10 (Keyring Connector<1>,Keyring<1>)
-  LimitAngle2 (Keyring<1>,Top Plate<1>)



Mass Properties:



Chaitanya - 9.1

Options...

Override Mass Properties...

Recalculate

☒ Include hidden bodies/components☐ Create Center of Mass feature☐ Show weld bead mass

Report coordinate values relative to: -- default --

Mass properties of Chaitanya - 9.1

Configuration: Default

Coordinate system: -- default --

Mass = 24.16 grams

Volume = 7360.17 cubic millimeters

Surface area = 7420.22 square millimeters

Center of mass: (millimeters)

X = 687.21

Y = 873.85

Z = 1276.31

Principal axes of inertia and principal moments of inertia: (grams * square millimeters)

Taken at the center of mass.

I_x = (0.97, 0.16, -0.19) P_x = 1426.01I_y = (-0.24, 0.38, -0.89) P_y = 9385.96I_z = (-0.07, 0.91, 0.41) P_z = 9814.50

Moments of inertia: (grams * square millimeters)

Taken at the center of mass and aligned with the output coordinate system. (Using positive tensor notation.)

L_{xx} = 1931.82 L_{xy} = 1283.56 L_{xz} = -1463.41L_{yx} = 1283.56 L_{yy} = 9529.21 L_{yz} = -407.96L_{zx} = -1463.41 L_{zy} = -407.96 L_{zz} = 9165.44

Moments of inertia: (grams * square millimeters)

Taken at the output coordinate system. (Using positive tensor notation.)

I_{xx} = 57798021.12 I_{xy} = 14507512.11 I_{xz} = 21185703.23I_{yx} = 14507512.11 I_{yy} = 50767230.46 I_{yz} = 26941304.09I_{zx} = 21185703.23 I_{zy} = 26941304.09 I_{zz} = 29863169.74